

Joint Gamma Schur Return

Exact integration, rational certificates and signed receiving bounds

14 September 2026

This volume contains the complete current joint-Schur note and the new continuation calculations, with the original packet, full confluent multiplicities, coefficient maps, mixed covariance and endpoint signs retained. Its accompanying source folder preserves the incoming edition and the full inherited Gamma, signed-return and original-degree proofs, together with their current receiving texts.

The incoming edition and the later calculations have distinct source records. Changes to the current note are recorded against the preserved incoming text. Display changes in this volume are recorded separately from mathematical source changes. The complete new proof bodies follow.

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The current joint-Schur note

Abstract

On the subsequence $k = 4l + 1$ we construct the exact Christoffel multiplication between the two original Gamma sources, calculate its joint arithmetic/boundary Schur complement, and return all four signs to $R_k^0 - R_k^\sigma$. The unconditioned boundary determinant is evaluated in a closed product with a proof using two explicit adjoint polynomial maps. Its signed contribution has limit $(\log 2)/8$ after division by q for a simple quartet; it is therefore not an error that can be discarded at the q scale. The remaining mixed contribution is specified by matrices of size l , with a convergent finite signed trace enclosure and the original relation-valued section correction. We also give the actual coefficient formula for every mixed plus/minus word and a complete rank classification for the first nontrivial $k = 5$, $(a, b) = (2, 3)$ case. All arithmetic Grams remain those of the given source; no offcritical zero is asserted to exist.

Current integration. This complete receiving edition preserves the supplied note and updates its actual arithmetic-error and continuation sites using the newer original-degree calculation. The archived original remains unchanged. The full companion proofs establish the factorial coordinate maps, GCR/GSR determinant identity, sharper free-boundary expansion, exact low-endpoint substitution, rational finite certificate and concrete PR31 resolvent return. These proofs accompany this text.

1 Scope, original input, and retained support

The input is the actual counterfactual packet stipulated in the continuation request:

$$h(s) = \prod_{\lambda \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \lambda)^m, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2. \quad (1)$$

Its full multiplicities and arithmetic function are retained:

$$g = 2\xi, \quad v_h = g/h, \quad U = M_{j_h v_h}, \quad w_h(y) = \frac{|v_h(\frac{1}{2} + iy)|^2}{2\pi}, \quad m_k = w_h^{*k}. \quad (2)$$

Fix $k = 4l + 1$, $l \geq 1$, put $c = k/2$, and set

$$e_k = 1 + k(m-1), \quad q = (k+1)^2 e_k, \quad \lambda_{ab} = c + (2a-k)\delta + i(2b-k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^{e_k}. \quad (3)$$

The ordered tensor fibre, all its multiplicities, and the weighted cyclic inclusion

$$E = \mathbb{C}[S]/(\chi) \xrightarrow{\eta} E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P \left(\sum_i s_i \right) 1, \quad v_h = j_h(g/h), \quad (4)$$

are unchanged. In particular, a determinant of this rectangular inclusion is never inverted.

The scalar and support observations remain

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad (p_R, b_R)(\tau) = (0, 0), \quad (p_R, b_R)(r^\bullet) = (r, 1). \quad (5)$$

The pair is injective. The structural map from the original pointed/preaddition base sends τ_0 to τ and 1 to $1_R^\bullet$. The arithmetic square retains its infinite integer target:

$$\begin{array}{ccc} G(\mathbb{Z}) & \longrightarrow & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \longrightarrow & \mathbb{C}. \end{array} \quad (6)$$

Every linear coefficient map constructed below is applied on each existing face and original outer leg. On a fixed present label ω its lift is $(\omega, x) \mapsto (\omega, Lx)$; external absence maps to external absence. On face inclusions j_A^B , zero insertion gives $L^B j_A^B = j_A^B L^A$. A represented zero, including an empty coefficient face at a present outer label, is not discarded. The existing intertype carrier maps and their nonunital sections retain their original types. These calculations concern their coefficient-linear arithmetic retracts; they do not assert that a linear section is a unital algebra homomorphism.

On the original distinguished tensor class the supplied same-class map is

$$I_k(v_k) = \frac{(k(m-1))!}{((m-1)!)^k} v_h(\rho)^k u_\rho^{\otimes k} \neq 0. \quad (7)$$

Multiplication by the comparison polynomial below carries this eigenclass to $f(k\rho)I_k(v_k)$, with $f(k\rho) \neq 0$; its explicitly constructed inverse restores it. On the surrounding primary module the complete triangular Leibniz matrix, not just this eigenvalue, is retained.

The original source remains

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int \phi = 0\}, \quad (8)$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad (9)$$

$$D = -x\partial_x, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x. \quad (10)$$

We use the existing Euler inverses only on their proved domains, $\mathcal{M}F_h = v_h$, and

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}_{h,k} = \eta \text{rem}_\chi. \quad (11)$$

All actual source norms still integrate $|P(c+iy)|^2 m_k(y) dy$. The Gamma norms used below are comparison norms on this same polynomial presentation.

Inherited determinant input. DMR/QDC, as stated in the new request, give $|\delta_k^\sigma| \leq 2\Xi_{k,2q}$ and the asserted rate for Ξ . This note does not reattribute that theorem or replace its full determinant proof by diagonal estimates. The supplied proof-file inventory is recorded separately. When a requested proof body is absent from that inventory, its theorem is an explicitly inherited input, not a claim of independent verification. All results from Section 2 through Section 7 below have their proofs here and do not assume the missing growing-family upper bound.

2 The exact two-reference multiplication

Retain

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad c_\sigma = \sqrt{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \quad r_k^0(y) = \frac{c_\sigma^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}. \quad (12)$$

Their masses are respectively c_σ and c_σ^k . Put $b_j = 2j + \frac{1}{2}$ and

$$F_l^\pm(y) = \prod_{j=0}^{l-1} (y \pm ib_j), \quad f_l(S) = \prod_{j=0}^{l-1} (S - \alpha_j), \quad \alpha_j = c + b_j = 2l + 2j + 1. \quad (13)$$

Gamma recurrence gives the holomorphic identity

$$\frac{\Gamma(l + 1/4 + iy/2)}{\Gamma(1/4 + iy/2)} = (i/2)^l F_l^-(y). \quad (14)$$

For real y , $F_l^- = \overline{F_l^+}$ and $F_l^+(y) = (-i)^l f_l(c + iy)$. Therefore

$$r_k^0(y) = C_k |f_l(c + iy)|^2 \sigma(y), \quad C_k = \frac{c_\sigma^k}{c_{k/4}} 4^{-l}. \quad (15)$$

The plus polynomial has not been substituted in the holomorphic recurrence: conjugation occurs at the displayed real-line observation. The alternative monic polynomial with roots $c - b_j$ is related by $f_l = (-1)^l (f_l^-)^\dagger c$, where $P^{\dagger c}(S) = \overline{P(2c - \bar{S})}$. Multiplication by f_l/f_l^- is the corresponding modulus-one map between their source images on this line. Its arithmetic action is $M_{f_l} M_{f_l^-}^{-1}$, not the identity.

For $k = 5$, $l = 1$, the complete formula is

$$f_1(S) = S - 3, \quad C_5 = 16\pi^2/3. \quad (16)$$

On the domain (3), every λ_{ab} has nonzero imaginary part because k is odd. Thus f_l and χ are coprime. The factor $M_f : E \rightarrow E$ is an E -module automorphism, with inverse the unique Bezout remainder. It satisfies

$$\eta M_f = M_{f(\sum s_i)} \eta. \quad (17)$$

All full-unit factors in (4) commute with this multiplication and stay in that equation.

2.1 Raw confluent jets and original remainders

Let $J_N : \mathcal{P}_N \rightarrow E$ be the monic remainder map. In raw derivative coordinates let

$$\mathcal{J}_\chi P = (P^{(r)}(\lambda_{ab}))_{a,b, 0 \leq r < e_k}. \quad (18)$$

The coordinate map $\mathcal{J}_\chi = \mathbf{E}_\chi J_N$ has entries

$$(\mathbf{E}_\chi)_{(\lambda,r),n} = \begin{cases} \frac{n!}{(n-r)!} \lambda^{n-r} & n \geq r, \\ 0 & n < r. \end{cases} \quad (19)$$

Its determinant is

$$\det \mathbf{E}_\chi = \left(\prod_\lambda \prod_{r=0}^{e_k-1} r! \right) \prod_{\lambda < \mu} (\mu - \lambda)^{e_k^2}, \quad (20)$$

for the stated lexicographic block order. Its inverse is the full Hermite interpolation map, not value-only interpolation. Multiplication by f has jet entries

$$(\mathbf{F}_\chi)_{r,s} = \binom{r}{s} f^{(r-s)}(\lambda) \quad (r \geq s) \quad (21)$$

inside each block and zero between blocks. Hence $M_f = \mathbf{E}_\chi^{-1} \mathbf{F}_\chi \mathbf{E}_\chi$ and $\det M_f = \prod_\lambda f(\lambda)^{e_k}$.

3 Joint full jets and the mixed Schur complement

Fix $N \geq q - 1$ and put $M = N + l$. All sources embed in the common $\mathcal{P}_{2q+l}$. Let H_M^σ be its unchanged sigma Gram. Let

$$Z_M : P \mapsto (P(\alpha_0), \dots, P(\alpha_{l-1})), \quad \mathcal{J}_M = \begin{pmatrix} J_M \\ Z_M \end{pmatrix}. \quad (22)$$

Coprimality and monic division prove that $\mathcal{J}_M$ is onto, with kernel $\chi f \mathcal{P}_{M-q-l}$. In particular its full joint covariance is positive definite:

$$\mathcal{J}_M (H_M^\sigma)^{-1} \mathcal{J}_M^* = \begin{pmatrix} K_M & C_M \\ C_M^* & A_M \end{pmatrix}. \quad (23)$$

Here $K_M = J_M(H_M^\sigma)^{-1}J_M^*$, $G_M^\sigma = K_M^{-1}$, and A_M is the covariance of the additional real nodes. Define both positive Schur complements

$$S_M = K_M - C_M A_M^{-1} C_M^*, \quad L_M = A_M - C_M^* G_M^\sigma C_M. \quad (24)$$

These formulas retain the actual cross block C_M .

Theorem 3.1 (Original tensor-Gamma quotient). *The canonical tensor-Gamma quotient is*

$$\boxed{G_N^0 = C_k M_f^* S_M^{-1} M_f}. \quad (25)$$

Its inverse and its finite-rank return are

$$(G_N^0)^{-1} = C_k^{-1} M_f^{-1} S_M M_f^{-*}, \quad (26)$$

$$G_N^0 = C_k M_f^* G_M^\sigma M_f + C_k (C_M^* G_M^\sigma M_f)^* L_M^{-1} (C_M^* G_M^\sigma M_f). \quad (27)$$

Proof. Multiplication $\mathbf{m}_f : \mathcal{P}_N \rightarrow \mathcal{P}_M$ is injective and has image $\ker Z_M$. Its source norm is exactly C_k^{-1} times the original tensor-Gamma norm by (15). The arithmetic constraint $J_N P = v$ is transported to $J_M(fP) = M_f v$. Thus its minimum is the minimum sigma norm with joint constraint $(M_f v, 0)$. Inverting (23) gives (25). Multiplication of the two block factors proves the Schur inverse identity in (27); it uses L_M , not a covariance with C_M erased. $\square$

The representative itself is

$$R_N^0 = \text{div}_f(H_M^\sigma)^{-1} \mathcal{J}_M^* \begin{pmatrix} K_M & C_M \\ C_M^* & A_M \end{pmatrix}^{-1} \begin{pmatrix} M_f \\ 0 \end{pmatrix}. \quad (28)$$

The division is defined on $\ker Z_M = f\mathcal{P}_N$. Both R_N^0 and R_N^σ satisfy $J_N R = 1$. With B_N the original multiplication-by- χ columns,

$$R_N^0 = R_N^\sigma - B_N (B_N^* H_N^0 B_N)^{-1} B_N^* H_N^0 R_N^\sigma. \quad (29)$$

Thus their difference is the original relation, with its actual coefficient. The source Pythagoras identity retains

$$(R_N^\sigma)^* H_N^0 R_N^\sigma - G_N^0 = (R_N^\sigma - R_N^0)^* H_N^0 (R_N^\sigma - R_N^0). \quad (30)$$

Applying $\mathcal{V}_{h,k}$ and the original fixed-order division by the factors $h(s_i)$ gives its theta primitive. The sign $(-1)^{i-1}$ in that primitive meets the same tensor-differential sign. At $N = q - 1$ the relation space is zero; every expression containing its projector is the zero map.

Equation (27) is an application of the already supplied selected-mode/Schur update: its base term is explicitly retained, and the update has rank at most l . It is not a new abstract Woodbury principle.

4 The exact signed return consists of small matrices

Let the four endpoints and signs be

$$(N_0, N_1, N_2, N_3) = (q - 1, q, 2q - 1, 2q), \quad (\varepsilon_0, \varepsilon_1, \varepsilon_2, \varepsilon_3) = (1, 1, -1, -1). \quad (31)$$

In the fixed monic sigma polynomial basis put

$$U_N = [Jp_{N+1}, \dots, Jp_{N+l}], \quad \Omega_N = \text{diag}(\omega_{N+1}^\sigma, \dots, \omega_{N+l}^\sigma), \quad X_N = \Omega_N^{-1} U_N^* G_N^\sigma U_N. \quad (32)$$

X_N is positive self-adjoint in Ω_N ; its entries retain all confluent derivatives through J . Define the second, differently metrized endomorphism

$$Y_N = A_{N+l}^{-1} C_{N+l}^* G_{N+l}^\sigma C_{N+l}. \quad (33)$$

It is self-adjoint in A_{N+l} and has spectrum in $[0, 1)$, since $L_{N+l} > 0$. The original kernel update gives

$$d_N := \log \frac{V_N^\sigma}{V_{N+l}^\sigma} = \log \det(I + X_N), \quad \beta_N := \log \frac{\det A_{N+l}}{\det L_{N+l}} = -\log \det(I - Y_N). \quad (34)$$

Taking the determinant of (25) yields

$$\log V_N^0 = q \log C_k + 2 \log |\det M_f| + \log V_{N+l}^\sigma + \beta_N. \quad (35)$$

The first two constants have not been removed before constructing these four metrics. Their signed sum is zero. In the exact notation of the incoming TWA return,

$$\boxed{R_k^0 - R_k^\sigma = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = \sum_{r=0}^3 \varepsilon_r \{ \log \det(I + X_{N_r}) + \log \det(I - Y_{N_r}) \}.} \quad (36)$$

The matrices in (36) have size $l = (k - 1)/4$, while the retained arithmetic module still has dimension q .

4.1 A positive determinant expression for the conditional block

Let $A_n^{\chi^\sigma}$ be the evaluation covariance at the same real nodes for polynomials of degree at most n in the original measure $|\chi(c + iy)|^2 \sigma(y) dy$. Orthogonal projection onto $\ker J_M = \chi \mathcal{P}_{M-q}$ proves

$$\boxed{L_M = \text{diag}(\chi(\alpha_j)) A_{M-q}^{\chi^\sigma} \text{diag}(\overline{\chi(\alpha_j)})}. \quad (37)$$

All $\chi(\alpha_j)$ are nonzero; more explicitly they are the positive products

$$\chi(\alpha_j) = \prod_{a=0}^k \prod_{b=0}^{(k-1)/2} ((\alpha_j - c - (2a - k)\delta)^2 + (2b - k)^2 \gamma^2)^{e_k}. \quad (38)$$

The common multiplier in (37) cancels only in the signed sum. At the first endpoint $N = q - 1$ one can also evaluate the joint determinant directly:

$$d_{q-1} - \beta_{q-1} = \log \frac{\left| \prod_j \chi(\alpha_j) \right|^2 \Delta_f^2}{\left(\prod_{n=q}^{q+l-1} \omega_n^\sigma \right) \det A_{q+l-1}}, \quad \Delta_f = 2^{l(l-1)/2} \prod_{r=1}^{l-1} r!. \quad (39)$$

Indeed the joint evaluation on the ordered basis $(1, \dots, S^{q-1}, \chi, \dots, \chi S^{l-1})$ has determinant $(\prod_j \chi(\alpha_j)) \Delta_f$, and the basis change has determinant one. No derivative factorial is lost: raw-jet coordinates introduce the same $\det E_\chi$ on the two sides and their equality was given explicitly above.

5 Closed evaluation of the free boundary determinant

The sigma monic polynomials satisfy, in $x = S - c$,

$$p_0 = 1, \quad p_1 = x, \quad p_{n+1} = x p_n + n(n - \tfrac{1}{2}) p_{n-1}, \quad \omega_n^\sigma = c_\sigma n! (\tfrac{1}{2})_n. \quad (40)$$

All masses remain in ω_n^σ . Write $\vartheta = 1/2$. At the real node α_j ,

$$p_n(\alpha_j) = (\vartheta)_n Q_j(n), \quad Q_j(n) = \sum_{r=0}^j \binom{j}{r} \frac{2^r n^r}{(\vartheta)_r}. \quad (41)$$

Proof. The exponential generating function derived from (40) solves $(1 - z^2)F' = (x + z/2)F$, $F(0) = 1$. At $x = 2j + 1/2$ it is $(1 + z)^j(1 - z)^{-j-1/2}$. Expanding the right side of (41), using $\sum_n (\vartheta)_n n^x z^n / n! = (\vartheta)_r z^r (1 - z)^{-\vartheta-r}$, gives the same series. Its leading coefficient as a polynomial in n is $2^j / (\vartheta)_j$. $\square$

It follows that

$$(A_M)_{ij} = \frac{1}{c_\sigma} \sum_{n=0}^M \frac{(\vartheta)_n}{n!} Q_i(n) Q_j(n). \quad (42)$$

This is an exact finite polynomial Gram, not an asymptotic kernel.

5.1 Proof of its monic norms with both measures retained

For any $\vartheta > 0$ define the discrete weights $w_n = (\vartheta)_n / n!$, $0 \leq n \leq M$, and the continuous measure

$$K_M t^{\vartheta-1} dt, \quad K_M = (\vartheta)_{M+1} / M!. \quad (43)$$

Their common mass is $(\vartheta)_{M+1} / (\vartheta M!)$. The joint measure

$$K_M \binom{M}{n} t^{n+\vartheta-1} (1-t)^{M-n} dt \quad (44)$$

has exactly these marginals by the beta integral. It defines the adjoint polynomial maps

$$(BP)(t) = \sum_{n=0}^M \binom{M}{n} t^n (1-t)^{M-n} P(n), \quad B^*(t^r)(n) = \frac{(n+\vartheta)_r}{(M+\vartheta+1)_r}. \quad (45)$$

The adjunction is in the two displayed original measures, with their common mass retained.

Let P_j be the monic continuous orthogonal polynomial. The formula

$$P_j(t) = \frac{(-1)^j}{(j+\vartheta)_j} t^{1-\vartheta} \frac{d^j}{dt^j} \{t^{j+\vartheta-1} (1-t)^j\} \quad (46)$$

has leading coefficient one. Integrating by parts j times proves orthogonality and

$$\|P_j\|^2 = K_M \frac{(j!)^2}{(2j+\vartheta)(j+\vartheta)_j^2}. \quad (47)$$

Every boundary term vanishes: at zero the last relevant derivative is $O(t^\vartheta)$, and at one the factor $(1-t)^j$ supplies the required zeros. Let Q_j^{mon} be the discrete monic polynomial. Adjointness and comparison of leading terms give

$$BQ_j^{\text{mon}} = M^j P_j, \quad B^* P_j = \frac{Q_j^{\text{mon}}}{(M+\vartheta+1)_j}. \quad (48)$$

Taking the same cross pairing in both measures proves its exact norm

$$\nu_j(M) = \frac{(\vartheta)_{M+j+1} (j!)^2}{(M-j)! (2j+\vartheta) (j+\vartheta)_j^2}. \quad (49)$$

This argument holds for every $0 \leq j \leq M$ and supplies all determinant factors.

Theorem 5.1 (Entire free boundary determinant). *For $M \geq l-1$ and $\vartheta = 1/2$,*

$$\det A_M = c_\sigma^{-l} \prod_{j=0}^{l-1} \frac{4^j (j!)^2 (\vartheta)_{M+j+1}}{(\vartheta)_j^2 (M-j)! (2j+\vartheta) (j+\vartheta)_j^2}. \quad (50)$$

Proof. In (42), changing the fixed monomial basis to Q_j has determinant $\prod_j 2^j/(\vartheta)_j$. The monic orthogonal-polynomial change has determinant one. Thus the Gram determinant is c_σ^{-l} times the square of the first determinant times $\prod_j \nu_j(M)$. Substitute (49). $\square$

In particular,

$$\frac{\det A_{M+1}}{\det A_M} = \prod_{j=0}^{l-1} \frac{M+j+\vartheta+1}{M+1-j}. \quad (51)$$

5.2 Its signed growing-family contribution is evaluated

Define

$$C_{q,l}^{\text{free}} = \log \frac{\det A_{2q+l-1} \det A_{2q+l}}{\det A_{q+l-1} \det A_{q+l}}, \quad B_l = l(l - \frac{1}{2}). \quad (52)$$

The expression is strictly positive. Formula (51) evaluates it by finite products. It also gives the uniform finite bounds

$$\boxed{2B_l \log \frac{2q+2l+1/2}{q+2l+1/2} \leq C_{q,l}^{\text{free}} \leq 2B_l \log 2.} \quad (53)$$

Proof. For $t_j = 2j + \vartheta$, each logarithmic increment is $\log(1 + t_j/(M+1-j))$. The inequalities $x/(1+x) \leq \log(1+x) \leq x$ apply with their positive denominators. Sum the two length- q windows in (51). Writing their indices as $n = q, \dots, 2q-1$, all lower-bound denominators are at most $n+2l+\vartheta$, and all upper-bound denominators are at least $n+1$. Since $\sum_j t_j = B_l$, integration of the decreasing functions $1/(n+c)$ gives (53). $\square$

The same bound gives the finite remainder

$$0 \leq 2B_l \log 2 - C_{q,l}^{\text{free}} \leq \frac{B_l(2l+1/2)}{q}. \quad (54)$$

Indeed the difference of the two logarithms is $\log(1 + (2l+1/2)/(2q+2l+1/2))$, at most $(2l+1/2)/(2q)$. For simple quartets, $q = (4l+2)^2$, so the squeeze gives

$$\boxed{\lim_{l \rightarrow \infty} \frac{C_{q,l}^{\text{free}}}{q} = \frac{\log 2}{8}.} \quad (55)$$

For each fixed $m > 1$, $q = (4l+2)^2(1 + (4l+1)(m-1))$, and the same ratio tends to zero. This is an evaluated component of the signed return, not a statement about the total mixed determinant.

Combining (36) and (37) gives the exact decomposition

$$\boxed{R_k^0 - R_k^\sigma = C_{q,l}^{\text{free}} + \sum_{r=0}^3 \varepsilon_r \{d_{N_r} + \log \det A_{N_r+l-q}^{\chi^\sigma}\}.} \quad (56)$$

The source/relation-dependent summand in (56) is retained. In particular (55) is not a limit for $R_k^0 - R_k^\sigma$ itself.

6 A finite signed enclosure and the exact return scale

Here is a finite enclosure for the entire expression (36), not just its free component. Use the common scalar

$$s = 2 + \sum_{r=0}^3 \text{Tr } X_{N_r}, \quad Z_N = I - (I + X_N)/s. \quad (57)$$

Each Z_N is positive self-adjoint in Ω_N with spectrum in $[0, 1]$; Y_N already has that property in A_{N+l} . Retaining the common determinant factor first gives

$$R_k^0 - R_k^\sigma = - \sum_r \varepsilon_r \sum_{n \geq 1} \frac{\text{Tr } Z_{N_r}^n + \text{Tr } Y_{N_r}^n}{n}. \quad (58)$$

The four $l \log s$ terms cancel before bounds are applied. There is no commutation assumption between different blocks.

For any one of these eight endomorphisms T , set

$$b_p(T) = \sum_{n=1}^p \frac{\text{Tr } T^n}{n}, \quad a_p(T) = \text{Tr } T^p, \quad u_p(T) = b_p(T) + \frac{b_{2p}(T) - b_p(T)}{1 - a_p(T)}. \quad (59)$$

For any p with $a_p(T) < 1$, the inherited positive-tail estimate gives $b_p(T) \leq -\log \det(I - T) \leq u_p(T)$. To check its applicability here, diagonalize only in the stated positive metric: all eigenvalues are in $[0, 1]$ by the proven Schur positivity. A finite such p therefore exists for each given matrix. Termwise $r_{2p}(x) \leq x^p r_p(x)$ proves the estimate without an additional arithmetic hypothesis.

Let $L_r = b_{p_r}(Z_{N_r}) + b_{q_r}(Y_{N_r})$ and $U_r = u_{p_r}(Z_{N_r}) + u_{q_r}(Y_{N_r})$, using any admitted orders. Then the entire signed return satisfies

$$\boxed{L_2 + L_3 - U_0 - U_1 \leq R_k^0 - R_k^\sigma \leq U_2 + U_3 - L_0 - L_1.} \quad (60)$$

Increasing the finite orders converges to the exact value. This is the existing PR29/STC trace certificate applied to the new small, fully specified joint-Schur matrices; it is not attributed as a new general logarithm inequality. All input errors must be included when numerical entries are used.

6.1 Return to the original arithmetic criterion

The original TWA equality remains

$$\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^0 + (R_k^0 - R_k^\sigma) + \delta_k^\sigma. \quad (61)$$

The complete original source deficits give the exact signed allowances

$$a_k^- := \Xi_{k,q-1} + \Xi_{k,q}, \quad a_k^+ := \Xi_{k,2q-1} + \Xi_{k,2q}, \quad -a_k^- \leq \delta_k^\sigma \leq a_k^+. \quad (62)$$

The sign proof is the original nonnegative quotient-deficit identity $\delta_k^\sigma = -b_{q-1} - b_q + b_{2q-1} + b_{2q}$ with $0 \leq b_N \leq \Xi_{k,N}$. It retains both low and both high cutoffs. The complete finite expressions JSR17–20 in the receiving proof give $\Xi_{k,N} \leq kq\Omega_{k,N}^{\text{deg}}$ and

$$D_k^- := kq(\Omega_{k,q-1}^{\text{deg}} + \Omega_{k,q}^{\text{deg}}), \quad D_k^+ := kq(\Omega_{k,2q-1}^{\text{deg}} + \Omega_{k,2q}^{\text{deg}}), \quad a_k^\pm \leq D_k^\pm. \quad (63)$$

Here $\Omega_{k,N}^{\text{deg}}$ is the explicitly displayed scalar original-degree majorant in JSR18. The positive matrix Ω_N retains its endpoint polynomial norms in the definition of X_N ; JSR24–25 combine the resulting Gamma certificate with the scalar majorant through the exact arithmetic receiving identity. All source integrals and constants of that majorant are given in JSR17, with the complete finite substitution proof following there.

If $[L_k, U_k]$ is the certificate (60), the current arithmetic receiver is

$$\boxed{\begin{aligned} \mathcal{B}_k^0 + L_k - a_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 + U_k + a_k^+, \\ \mathcal{B}_k^0 + L_k - D_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 + U_k + D_k^+. \end{aligned}} \quad (64)$$

The first interval is contained in the previous symmetric interval because each $a_k^\pm \leq 2\Xi_{k,2q}$. The second evaluates its allowances using the full finite source estimate. The exact two low Gamma

endpoints in JSA26–30 can be substituted before bounding; this removes precisely their two truncation widths from the original certificate with the same high-endpoint orders. All original mixed covariance factors remain in that narrower interval.

The original amplification identities give, on either length- q window,

$$L_{h,k}^{2q} \leq \frac{\omega_j^{\text{ar}}}{\omega_i^{\text{ar}}} \frac{V_i^{\text{ar}}}{V_j^{\text{ar}}}, \quad L_{h,k} \geq \frac{\delta}{2} kq, \quad (65)$$

with the separate first-admissible-degree formula at $i = q - 1$. The original HC5 lower bound remains $4q \log(D_h k)$ with the same D_h . In particular the retained residual satisfies

$$\boxed{\begin{aligned} \max\{0, \mathcal{B}_k^0 + L_k - D_k^- - 4q \log(D_h k)\} &\leq \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k) \\ &\leq \mathcal{B}_k^0 + U_k + D_k^+ - 4q \log(D_h k). \end{aligned}} \quad (66)$$

This is the updated upper slack with the full Gamma return.

The original-degree estimate, with $p = 8m - 9/2$ and $\eta = (p - 1)/p$, is

$$\frac{\Xi_{k,N}}{kq} = O_h\left(\frac{\log k}{k} + \left(\frac{k \log k}{q}\right)^\eta\right) \quad (N = q - 1, q, 2q - 1, 2q). \quad (67)$$

For $m \geq 2$, retaining $q = (k + 1)^2[1 + k(m - 1)]$ gives the precise signed limits of the finite majorants

$$\frac{D_k^-}{q \log k} \longrightarrow 64m + 245, \quad \frac{D_k^+}{q \log k} \longrightarrow 128m + 490. \quad (68)$$

Their proof is JSR18–21, including every finite N factor. For $m = 1$, the current source bound instead retains $\Xi_{k,N}/(kq) = O_h((\log k/k)^{5/7})$. These statements replace the older rate at this receiving site; they do not identify the signed Gamma term with the arithmetic defect.

A log-volume error E_k changes a window exponential by factors between $e^{-E_k/(2q)}$ and $e^{E_k/(2q)}$. Thus $o(q)$ gives a factor tending to one and $o(q \log k)$ retains the leading logarithmic threshold. The free summand has the sharper proved finite expansion

$$C_{q,l}^{\text{free}} = 2l(l - \tfrac{1}{2}) \log 2 - \frac{l(l - \tfrac{1}{2})(l + 1/4)}{q} + R_{q,l}, \quad 0 \leq R_{q,l} \leq \frac{l(l - \tfrac{1}{2})(14l^2 + 5l + 3)}{16q^2}. \quad (69)$$

The complete free-boundary proof below gives a second-order enclosure and its constant terms at the same original degrees. Its nonzero q -scale limit for simple quartets remains part of the exact return. The remaining growing-family calculation concerns the relation-dependent Schur term together with the original arithmetic deficit in the displayed upper slack. Fixed-input convergence and the free summand alone do not supply a sign for that entire expression.

7 Every mixed word and its actual observation

This section works on the reduced pair observations of the original full quartet, not on a new assertion that every original root is simple. All higher jets remain in the kernel of the reduced maps. Put

$$c_\pm = \tfrac{1}{2} \pm \delta, \quad h_\pm(s) = (s - c_\pm)^2 + \gamma^2. \quad (70)$$

The reduced remainder from the full packet is onto each indicated value algebra. The reduced maps are not used as faithful observations of the full packet: for the original top local eigenvector v_k at $k\rho$, every reduced value map kills it when $e_k > 1$. When $e_k = 1$, the word map to real-margin a kills it unless $a = k$. These claims follow by reducing its local representative $(S - k\rho)^{e_k - 1}$ and by the distinct real margins in (3). The actual nonzero class remains in the specified kernel before that quotient. The source continues to use v_h , not the different literal divisor function $v_{h_\pm} = h/h_\pm v_h$.

Let $\Pi_{\pm}(u, 0)$ be the convergent oriented pair period matrices in their monomial frames, and let W be the retained invertible character-coordinate matrix of the cubic cover. Define

$$E_{\pm} = \begin{pmatrix} 1 & c_{\pm} - i\gamma \\ 1 & c_{\pm} + i\gamma \end{pmatrix}, \quad U_{\pm} = \text{diag}(v_h(c_{\pm} - i\gamma), v_h(c_{\pm} + i\gamma)), \quad T_{\pm} = W^{-1}\Pi_{\pm}E_{\pm}^{-1}U_{\pm}. \quad (71)$$

The determinant of $E_{\pm}$ is $2i\gamma$. Both $U_{\pm}$ are invertible at the marked actual quartet, and every entry is retained. These maps go from original reduced jet values, through weighted polynomial coefficients, to actual character period coordinates.

For a word with a plus and $b = k - a$ minus factors, use the ordered unscaled symmetric orbit vectors. Their coordinate Gram entries are $\binom{a}{i}\binom{b}{j}$; the actual period and theta forms are still obtained by their displayed pullbacks, not assigned this coordinate Gram. Grouping an ordered word uses its explicit permutation and the top-cochain Koszul sign.

Write the two rows of T_+ as linear forms $A(X, Y), B(X, Y)$, and those of T_- as $C(X, Y), D(X, Y)$. The selected rows are

$$\mathcal{I}_{a,b} = \{(i, j) : 0 \leq i \leq a, 0 \leq j \leq b, k + i + j \equiv 0 \pmod{3}\}. \quad (72)$$

For each such row expand the actual degree- k form

$$A^{a-i}B^iC^{b-j}D^j = \sum_{r=0}^k \mathbf{C}_{(i,j),r} X^{k-r}Y^r. \quad (73)$$

These are explicit finite binomial sums in the entries of (71); no rank is assigned from the row count. Put

$$\lambda_r = ac_+ + bc_- + (2r - k)i\gamma, \quad \mathbf{E}_{a,b} = (\lambda_r^n)_{0 \leq r, n \leq k}. \quad (74)$$

The roots are distinct, and $\det \mathbf{E}_{a,b} = (2i\gamma)^{k(k+1)/2} \prod_{j=1}^k j!$. The exact selected-row map on the cyclic reduced coefficient algebra is

$$\boxed{\Lambda_{a,b} = \mathbf{C}\mathbf{E}_{a,b}} \quad (75)$$

Proof. In an unscaled orbit sum, the coefficient of a fixed output word is the product of its row linear forms. Summing all input words with exactly r upper-root entries extracts the coefficient of $X^{k-r}Y^r$. The sum operator on that input has eigenvalue λ_r . Evaluation of the cyclic polynomial gives precisely (75). This also proves that no extra inverse binomial coefficient belongs in the formula. $\square$

For arbitrary a, b its exact rank is determined by this finite coefficient matrix, with no maximal-rank presumption. One exact algebraic implementation forms the coefficients of $\det(I + t\mathbf{C}^*H\mathbf{C})$ with the retained positive row Gram H ; the largest nonzero coefficient is the rank. A lexicographically first nonzero minor supplies a basis of the image and, by solving the retained pivot columns against the nonpivot columns, the complete kernel. Multiplication by $\mathbf{E}_{a,b}^{-1}$ returns that kernel to the original cyclic coefficient frame. These are finite coefficient calculations for the actual period entries, not an assumption that an unevaluated minor is nonzero.

7.1 Completed first odd-subsequence mixed case: $k = 5, a = 2, b = 3$

The selected rows are $(0, 1), (1, 0), (1, 3), (2, 2)$. Their polynomials are

$$A^2C^2D, \quad ABC^3, \quad ABD^3, \quad B^2CD^2. \quad (76)$$

The invertible coordinate map $(X, Y) \mapsto (A, B)$ and its induced Sym^5 map have their full inverse from T_+^{-1} . In those coordinates write

$$C = \alpha A + \beta B, \quad D = \gamma_0 A + \delta_0 B, \quad \begin{pmatrix} \alpha & \beta \\ \gamma_0 & \delta_0 \end{pmatrix} = T_- T_+^{-1}. \quad (77)$$

Theorem 7.1 (True rank, including the exceptional locus). *The rank of (76) is two when $\beta = \gamma_0 = 0$, and four otherwise. The reduced cyclic kernel consequently has dimension four or two, respectively. Its full-quartet preimage has dimension $q - 2$ or $q - 4$, retaining all omitted roots and all higher jets.*

Proof. When $C = \alpha A, D = \delta_0 B$, the first two rows are nonzero multiples of $A^4 B$ and the last two are multiples of AB^4 , proving rank two. If all four coefficients are nonzero, evaluate a linear dependence at $A = 0$ to kill the fourth coefficient and at $B = 0$ to kill the first. The remaining dependence is $AB(uC^3 + vD^3) = 0$, and independent C, D force $u = v = 0$. It remains to check the possible zero coefficients of an invertible two-by-two matrix. If $\beta = 0, \gamma_0 \neq 0$, evaluation at $B = 0$ first kills the first row coefficient. After division by the common nonzero polynomial AB , evaluation at $D = 0$ kills the second, and the coefficients of A, B kill the other two. The case $\gamma_0 = 0, \beta \neq 0$ is the reversed argument: evaluate at $A = 0$, divide by AB , then evaluate at $C = 0$. If $\alpha = 0, \delta_0 \neq 0$, evaluation at $A = 0$ kills the fourth coefficient; after division by AB , evaluation at $B = 0$ kills the third and the remaining two are independent because $\gamma_0 \neq 0$. If $\delta_0 = 0, \alpha \neq 0$, reverse that argument. Finally $\alpha = \delta_0 = 0$ gives four distinct monomials $A^3 B^2, AB^4, A^4 B, A^2 B^3$. These cases exhaust every invertible matrix. $\square$

For rank two the kernel in the transformed binary coefficient frame is explicitly the annihilator of the $A^4 B$ and AB^4 coefficients. For rank four take the four rows in their stated order; any first nonzero four-column minor gives the exact two free-column kernel vectors by its matrix inverse. This returns through the explicit $\text{Sym}^5 T_+$ and $E_{2,3}$ maps; no auxiliary metric changes the kernel.

7.2 The two potentials' discrepancy is calculated

The translation $s = c_{\pm} + z$ has Jacobian one and preserves the oriented contour differences. Thus

$$\Pi_{\pm} E_{\pm}^{-1} = e^{C_{\pm}/u} \Pi_{\text{cen}} E_0^{-1}, \quad C_{\pm} = c_{\pm}^3/3 + c_{\pm} \gamma^2. \quad (78)$$

Put $B_0 = W^{-1} \Pi_{\text{cen}} E_0^{-1}$ and $z_0 = v_h(\rho) \neq 0$. Reflection and conjugation of the original even-degree quartet give

$$U_+ = \text{diag}(\bar{z}_0, z_0), \quad U_- = \text{diag}(z_0, \bar{z}_0), \quad T_- T_+^{-1} = e^{(C_- - C_+)/u} B_0 \text{diag}(z_0/\bar{z}_0, \bar{z}_0/z_0) B_0^{-1}. \quad (79)$$

This is the exact discrepancy entering the rank theorem. For $B_0 = \begin{pmatrix} a_0 & b_0 \\ c_0 & d_0 \end{pmatrix}$ the offdiagonal entries are, respectively,

$$\frac{a_0 b_0 (\bar{z}_0/z_0 - z_0/\bar{z}_0)}{\det B_0}, \quad \frac{c_0 d_0 (z_0/\bar{z}_0 - \bar{z}_0/z_0)}{\det B_0}, \quad (80)$$

with the same retained exponential scalar. Consequently the exceptional rank-two locus is exactly the vanishing of both these expressions. No phase condition on the actual z_0 is assumed.

8 Joint boundary covariance and the same arithmetic class

Let $p_{a,b} : E \rightarrow E_{a,b}$ be the actual reduced remainder. Its kernel retains every other primary block and the higher jets in the selected blocks. Its metric is the induced original quotient

$$G_{a,b} = (p_{a,b} G^{-1} p_{a,b}^*)^{-1}, \quad r_{a,b} = G^{-1} p_{a,b}^* G_{a,b}. \quad (81)$$

Then $p_{a,b} r_{a,b} = 1$ and $1 - r_{a,b} p_{a,b}$ projects onto its full kernel in the original G metric.

Combine the existing observation Λ and the calculated word map by

$$\mathcal{O} = \begin{pmatrix} \Lambda \\ \Lambda_{a,b} p_{a,b} \end{pmatrix} : E \rightarrow \text{im } \mathcal{O}. \quad (82)$$

On its actual image the covariance and minimum section are

$$\Sigma = \mathcal{O}G^{-1}\mathcal{O}^*, \quad s = G^{-1}\mathcal{O}^*\Sigma^{-1}, \quad \ker \mathcal{O} = \ker \Lambda \cap p_{a,b}^{-1}(\ker \Lambda_{a,b}). \quad (83)$$

In the larger product target the covariance has all four blocks

$$\begin{pmatrix} \Lambda G^{-1}\Lambda^* & \Lambda G^{-1}p_{a,b}^*\Lambda_{a,b}^* \\ \Lambda_{a,b}p_{a,b}G^{-1}\Lambda^* & \Lambda_{a,b}p_{a,b}G^{-1}p_{a,b}^*\Lambda_{a,b}^* \end{pmatrix}. \quad (84)$$

Its inverse is taken only on $\text{im } \mathcal{O}$. The conditional Schur complement may have a kernel and that kernel is retained; no independent-observation assumption removes either cross term. Substituting (25) or its inverse computes this covariance for the exact tensor-Gamma return. Substituting the original arithmetic G computes it for the unchanged theta source.

At general roots ζ of $h - t$, if $v_h(\zeta + z) = z^\nu a(z)$ with a a unit, multiplication on $\mathbb{C}[z]/z^r$ has image $(z^{\min(\nu, r)})$ and kernel $(z^{r-\min(\nu, r)})$. This is the explicit extension of the marked unit map used above; it is not an unproved invertibility claim away from the marked fibre. The source Fourier signs, oriented period contours, and nilpotent exponentials remain those of the corrected transfer. In particular original dilation is still the ordered finite exponential, not merely the semisimple eigenvalue.

9 What the calculation establishes

The new result is the exact remaining Gamma comparison on the unbounded subsequence, its original source/quotient maps, a closed evaluation of the free extra-node determinant, the finite signed mixed-Schur certificate, and a non-maximal-rank classification for the first mixed word required on that subsequence. The term in (55) has an actual nonzero q -scale limit for simple quartets. The mixed term in (56), however, remains necessary; neither its sign nor its growing-family cancellation has been supplied by the free determinant alone.

The unresolved expression after these calculations is explicit: the signed sum of $\log \det(I + X_N)$ and $\log \det(I - Y_N)$ in (36), or equivalently the mixed term in (56), combined with the inherited actual arithmetic δ_k^q . Formula (60) is a terminating fixed-input enclosure, not a tensor-uniform estimate. Formula (64) is its exact return to the original contradiction criterion. No claim of an RH contradiction, an actual counterexample, or failure of the absolute-base programme is made.

Evidence and source record. The accompanying scripts test the finite polynomial and covariance identities on explicitly declared exact calibrations. They do not assert that the calibration roots divide 2ξ . The packet assumptions, DMR/QDC estimates, TWA/BRU/HC and existing STC/PR29 results retain their supplied scope. New execution logs, archive inventories and PDF-build receipts are separate from inherited records. Incoming archives and standalone files are retained byte-for-byte in the cumulative delivery. The new note has a complete independent proof body; retained historical source files are not silently edited.

Exact integration with the full-jet and relation determinants

This calculation supplies the exact coordinate morphisms between the incoming joint-Schur formula and the existing GCR and GSR formulas. It then uses the already exactly computable low endpoints to remove four unnecessary trace approximations from the incoming certificate. All degrees, source masses, full confluent multiplicities and mixed covariance factors remain present. No statement below asserts the existence of a zero of ξ outside the critical line.

10 Original finite source and its orthogonal polynomials

Fix integers $l \geq 1$, $m \geq 1$ and real parameters $0 < \delta < 1/2$, $\gamma > 2$. Retain

$$\begin{aligned} k &= 4l + 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \\ \lambda_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^e, \\ t_j &= 2j + \frac{1}{2}, \quad \alpha_j = c + t_j, \quad f(S) = \prod_{j=0}^{l-1} (S - \alpha_j), \quad E = \mathbb{C}[S]/(\chi), \quad \mathcal{P}_N = \mathbb{C}[S]_{\leq N}. \end{aligned} \tag{JSA1}$$

Negative-degree polynomial spaces are zero. All quotient coordinates on E are the original increasing monomials $1, S, \dots, S^{q-1}$. Let $J_N : \mathcal{P}_N \rightarrow E$ be monic remainder. For $N \geq q - 1$ it is onto and has kernel $\chi \mathcal{P}_{N-q}$: polynomial division proves both assertions, including the zero kernel at $N = q - 1$.

The two unchanged measures on the original line $S = c + iy$ are

$$\begin{aligned} d\sigma(y) &= \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad c_\sigma = \sqrt{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \\ dm_0(y) &= \frac{c_\sigma^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi} dy, \quad C_k = \frac{c_\sigma^k}{4^l c_{k/4}} = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}. \end{aligned} \tag{JSA2}$$

Gamma recurrence, applied before taking an absolute value, gives

$$\frac{\Gamma(l + 1/4 + iy/2)}{\Gamma(1/4 + iy/2)} = (i/2)^l \prod_{j=0}^{l-1} (y - it_j).$$

For real y , $f(c + iy) = i^l \prod_j (y + it_j)$, so

$$dm_0 = C_k |f(c + iy)|^2 d\sigma. \tag{JSA3}$$

The conjugate products have been related only on this real integration line. Multiplication $T_{f,N} : \mathcal{P}_N \rightarrow \mathcal{P}_{N+l}$ is injective and satisfies $H_N^0 = C_k T_{f,N}^* H_{N+l}^\sigma T_{f,N}$ in the increasing monomial frames. Here H always denotes the actual Gram for its stated measure.

For completeness the original mass and polynomial norms can be calculated directly. For $a > 0$, the Euler integrals for the two Gamma factors, with $u + v = w$ and $r = \log(u/v)$, have Jacobian $we^r/(1 + e^r)^2$. Integrating w therefore gives

$$|\Gamma(a + iy/2)|^2 = 2\Gamma(2a) \int_{\mathbb{R}} e^{iyt} (2 \cosh t)^{-2a} dt.$$

Fourier inversion gives the characteristic function

$$\int_{\mathbb{R}} e^{isy} \frac{|\Gamma(a + iy/2)|^2}{2\pi} dy = 2\Gamma(2a) (2 \cosh s)^{-2a}. \tag{JSA4}$$

These integrals are absolutely convergent; the integrand on the right before Fourier inversion is smooth with exponentially decaying derivatives, so the usual Fourier inversion follows also by an integrable Gaussian regularization and dominated convergence. At $a = 1/4$ this is $c_\sigma(\cosh s)^{-1/2}$.

Define monic p_n by

$$\sum_{n \geq 0} p_n(S) \frac{z^n}{n!} = (1 - z^2)^{-1/4} \exp((S - c) \operatorname{arctanh} z). \quad (\text{JSA5})$$

For real z, w sufficiently close to zero, JSA4 and $\cosh(\operatorname{arctanh} z - \operatorname{arctanh} w) = (1 - zw)/\sqrt{(1 - z^2)(1 - w^2)}$ show that the integral of the first generating function times the conjugate second generating function is $c_\sigma(1 - zw)^{-1/2}$. Differentiation under the integral is justified by the exponential Gamma decay and the bounded parameters. Comparing coefficients proves

$$\langle p_n, p_r \rangle_\sigma = \mathbf{1}_{n=r} \omega_n, \quad \omega_n = c_\sigma n! (1/2)_n, \quad p_{n+1} = (S - c)p_n + n(n - 1/2)p_{n-1}. \quad (\text{JSA6})$$

The recurrence also follows by differentiating JSA5. For general $a > 0$ the generating function $(1 - z^2)^{-a} \exp((S - c) \operatorname{arctanh} z)$ has integrated product $c_a(1 - zw)^{-2a}$ by JSA4. Its degree- n monic norm is therefore $c_a n! (2a)_n$. Taking $a = k/4$ and multiplying by the original factor $c_\sigma^k / c_{k/4}$ gives the monic norms for dm_0 ,

$$\omega_n^0 = c_\sigma^k n! \frac{\Gamma(n + k/2)}{\Gamma(k/2)}.$$

Every monic triangular change has determinant one. Hence

$$\frac{\det H_N^f}{\det H_N^\sigma} = D_{l,N} := \prod_{a=0}^N \prod_{r=0}^{2l-1} (a + r + 1/2), \quad H_N^f = T_{f,N}^* H_{N+l}^\sigma T_{f,N}. \quad (\text{JSA7})$$

Indeed $H_N^0 = C_k H_N^f$ and $\omega_a^0 / (C_k \omega_a) = \Gamma(a + 2l + 1/2) / \Gamma(a + 1/2)$, which proves every displayed factor.

11 The exact factorial morphism on all primary jets

Order the pairs (a, b) lexicographically and the derivatives within each root increasingly. Define

$$\mathcal{E}_{(\lambda, r), n}^d = \begin{cases} \binom{n}{r} \lambda^{n-r} & n \geq r, \\ 0 & n < r, \end{cases} \quad D_\chi = \operatorname{diag}_{\lambda, 0 \leq r < e} (r!), \quad \mathcal{E}^r = D_\chi \mathcal{E}^d. \quad (\text{JSA8})$$

Here $\mathcal{E}^d : E \rightarrow \bigoplus_\lambda \mathbb{C}^e$ is the full divided-derivative evaluation, and $\mathcal{E}^r$ is the full raw-derivative evaluation. Both are invertible: vanishing of all derivatives through order $e - 1$ at every root means divisibility by the degree- q polynomial χ , and a representative of degree less than q must then be zero. Since their source and target dimensions are q , injectivity proves surjectivity. The exact isomorphism of their targets is D_χ , with inverse $\operatorname{diag}(1/r!)$; no derivative order is discarded.

For multiplication by f the two block matrices are

$$(F_\chi^d)_{rs} = f^{(r-s)}(\lambda) / (r - s)!, \quad (F_\chi^r)_{rs} = \binom{r}{s} f^{(r-s)}(\lambda) \quad (r \geq s), \quad (\text{JSA9})$$

and zero above the diagonal or between different roots. The Taylor coefficient multiplication and the raw Leibniz rule prove these formulas directly. In particular

$$F^r = D_\chi F^d D_\chi^{-1}, \quad M_f = (\mathcal{E}^d)^{-1} F^d \mathcal{E}^d = (\mathcal{E}^r)^{-1} F^r \mathcal{E}^r, \quad \det M_f = \prod_{a,b} f(\lambda_{ab})^e. \quad (\text{JSA10})$$

The imaginary parts $(2b - k)\gamma$ are nonzero because k is odd, whereas all α_j are real. Every diagonal factor in JSA10 is therefore nonzero. The inverse of M_f is the remainder of the Bezout inverse

of f modulo χ . This same map obeys $\eta M_f = M_{f(\sum s_i)} \eta$ on the original weighted cyclic inclusion: evaluate the polynomial in the commuting sum operator and retain the commuting original factor $v_h^{\otimes k}$.

For any onto observation A from a source with Gram $H > 0$, write $K = AH^{-1}A^*$. A target-coordinate change T produces

$$K' = TKT^*, \quad G' = (K')^{-1} = T^{-*}K^{-1}T^{-1}, \quad H^{-1}(TA)^*G'T = H^{-1}A^*K^{-1}. \quad (\text{JSA11})$$

These identities follow by multiplying the displayed inverses. Thus the minimum source representative is the same vector after the exact target-coordinate isomorphism. Applied to JSA8, the joint covariance transforms by $\text{diag}(D_\chi, I_l)$, and both its arithmetic Schur complement and its inverse transform by the corresponding congruences. In particular raw jets introduce no hidden factorial into the original quotient metric.

12 Joint covariance and the complete GCR cross factors

Fix $N \geq q - 1$ and $M = N + l$. Define

$$Z_M : P \mapsto (P(\alpha_0), \dots, P(\alpha_{l-1})), \quad \mathcal{J}_M = \begin{pmatrix} J_M \\ Z_M \end{pmatrix}, \quad \mathcal{J}_M(H_M^\sigma)^{-1} \mathcal{J}_M^* = \begin{pmatrix} K_M & C_M \\ C_M^* & A_M \end{pmatrix}. \quad (\text{JSA12})$$

Interpolation modulo the coprime product χf proves that $\mathcal{J}_M$ is onto and has kernel $\chi f \mathcal{P}_{M-q-l}$. Its covariance is positive definite: a nonzero target covector pulls back to a nonzero source covector by surjectivity, and $H_M^{-1} > 0$. Completing squares proves positivity of both

$$S_M = K_M - C_M A_M^{-1} C_M^*, \quad L_M = A_M - C_M^* K_M^{-1} C_M. \quad (\text{JSA13})$$

The source minimum for $Ax = v$ is $H^{-1}A^*(AH^{-1}A^*)^{-1}v$: it has observation v and is H -orthogonal to $\ker A$, which proves its unique minimum property by expansion of the squared norm. The map $T_{f,N}$ identifies the source constraint $J_N P = v$ with the joint constraint $(J_M(fP), Z_M(fP)) = (M_f v, 0)$. Conversely the latter constraint implies $f \mid Q$ and $Q = fP$ for a unique $P \in \mathcal{P}_N$; invertibility of M_f then gives $J_N P = v$. Consequently the exact original quotient and its section are

$$G_N^0 = C_k M_f^* S_M^{-1} M_f, \quad R_N^0 = T_{f,N}^{-1} (H_M^\sigma)^{-1} \mathcal{J}_M^* \begin{pmatrix} K_M & C_M \\ C_M^* & A_M \end{pmatrix}^{-1} \begin{pmatrix} M_f \\ 0 \end{pmatrix}. \quad (\text{JSA14})$$

The inverse $T_{f,N}^{-1}$ is used only on $\ker Z_M = f \mathcal{P}_N$. Direct block multiplication yields

$$G_N^0 = C_k M_f^* K_M^{-1} M_f + C_k (C_M^* K_M^{-1} M_f)^* L_M^{-1} (C_M^* K_M^{-1} M_f). \quad (\text{JSA15})$$

In particular the finite update retains the actual C_M on both sides of the conditional inverse.

To identify every GCR factor, set

$$U = [Jp_{N+1}, \dots, Jp_{N+l}], \quad \Omega = \text{diag}(\omega_{N+1}, \dots, \omega_{N+l}), \quad G_N^\sigma = K_N^{-1}, \quad (\text{JSA16}) \\ X = \Omega^{-1} U^* G_N^\sigma U, \quad Y = A_M^{-1} C_M^* K_M^{-1} C_M, \quad K_M = K_N + U \Omega^{-1} U^*.$$

The last equality is exactly the difference of the two finite orthogonal-polynomial covariance sums. Put $E_d = \mathcal{E}^d$. The GCR divided-jet matrices, with their original meanings, are

$$\mathcal{K}_N = E_d K_N E_d^*, \quad \mathcal{U} = E_d U \Omega^{-1/2}, \quad \mathcal{V} = E_d C_M A_M^{-1/2}, \quad \mathcal{S} = E_d S_M E_d^*. \quad (\text{JSA17})$$

Thus GCR's $K + \mathcal{U} \mathcal{U}^* - \mathcal{V} \mathcal{V}^*$ is literally $\mathcal{S}$. Its two positive factors satisfy

$$\mathcal{A} = I + \mathcal{U}^* \mathcal{K}_N^{-1} \mathcal{U} = \Omega^{1/2} (I + X) \Omega^{-1/2}, \quad (\text{JSA18}) \\ \mathcal{B} = I - \mathcal{V}^* (\mathcal{K}_N + \mathcal{U} \mathcal{U}^*)^{-1} \mathcal{V} = A_M^{1/2} (I - Y) A_M^{-1/2}, \\ \mathcal{B} = I - \mathcal{V}^* \mathcal{K}_N^{-1} \mathcal{V} + \mathcal{V}^* \mathcal{K}_N^{-1} \mathcal{U} \mathcal{A}^{-1} \mathcal{U}^* \mathcal{K}_N^{-1} \mathcal{V}.$$

The last formula follows by multiplying out the inverse of $\mathcal{K}_N + \mathcal{U}\mathcal{U}^*$; the preceding equality proves that it is exactly the incoming mixed conditional factor. It contains the full cross product of the new-degree columns and the extra-node columns. Positivity follows from JSA13. Consequently X is positive self-adjoint in Ω , Y is positive self-adjoint in A_M , and its eigenvalues are strictly less than one. These are metric similarities, not assertions that their raw matrices are Hermitian in an unspecified metric.

Write $\mathcal{W} = (\mathcal{U} \mathcal{V})$ and $\mathcal{J} = \text{diag}(I_l, -I_l)$. Computing a block determinant in either order proves $\det(I + AB) = \det(I + BA)$, even for rectangular A, B . Applied here it proves

$$\mathfrak{d}_N := \frac{\det S_M}{\det K_N} = \det(I_{2l} + \mathcal{J}\mathcal{W}^* \mathcal{K}_N^{-1} \mathcal{W}) = \det(I + X) \det(I - Y) > 0. \quad (\text{JSA19})$$

In particular the incoming notation $d_N = \log \det(I + X)$ is the logarithm of the degree-increase factor alone, whereas the GCR notation d_N is the entire positive number $\mathfrak{d}_N$. Their exact relation is $\log \mathfrak{d}_N = d_N - \beta_N$ with $\beta_N = -\log \det(I - Y)$.

13 Original relation map and the signed endpoint equality

Let B_N consist of the original columns $\chi, \chi S, \dots, \chi S^{N-q}$. When $N = q - 1$ it has no columns. The minimum sections satisfy

$$\begin{aligned} R_N^0 &= R_N^\sigma - B_N(B_N^* H_N^0 B_N)^{-1} B_N^* H_N^0 R_N^\sigma, \\ (R_N^\sigma)^* H_N^0 R_N^\sigma - G_N^0 &= (R_N^\sigma - R_N^0)^* H_N^0 (R_N^\sigma - R_N^0). \end{aligned} \quad (\text{JSA20})$$

Indeed the first vector has the same remainder and is H_N^0 -orthogonal to every column of B_N , so it is the unique minimum. Expanding the square and using that orthogonality proves the second identity. At the empty relation space both correction expressions are zero.

For $r_N = N + 1 - q$, retain the full relation determinant

$$\mathfrak{R}_N = \frac{\det \text{Gram}_\sigma(f\chi, f\chi S, \dots, f\chi S^{r_N-1})}{\det \text{Gram}_\sigma(\chi, \chi S, \dots, \chi S^{r_N-1})}. \quad (\text{JSA21})$$

Both empty determinants are one. The frame $(1, \dots, S^{q-1}, \chi, \dots, \chi S^{N-q})$ has determinant one in increasing monomial coordinates because χ is monic. In this frame elimination of the relation cross block gives $\det H = \det(B_N^* H B_N) \det G$. The quotient block is G by the proved minimum property. Apply the identity to H_N^f and H_N^σ and then use JSA7 and JSA14. This gives

$$\log \mathfrak{d}_N = 2 \log |\det M_f| - \log D_{l,N} + \log \mathfrak{R}_N. \quad (\text{JSA22})$$

Thus the two l -dimensional determinants, the GCR $2l$ determinant, and the GSR determinant on the entire original relation subspace have exactly the same endpoint value under JSA22.

Set $N_0 = q - 1, N_1 = q, N_2 = 2q - 1, N_3 = 2q$ and $(\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) = (1, 1, -1, -1)$. Taking determinants in JSA14 first gives

$$\log \det G_N^0 - \log \det G_N^\sigma = q \log C_k + 2 \log |\det M_f| - \log \mathfrak{d}_N. \quad (\text{JSA23})$$

Only after constructing all four endpoints do the first two terms cancel. If $\mathfrak{T} = \sum_r \epsilon_r (\log \det G_{N_r}^0 - \log \det G_{N_r}^\sigma)$, the incoming return is consequently

$$R_k^0 - R_k^\sigma = -\mathfrak{T} = \sum_{r=0}^3 \epsilon_r \log \mathfrak{d}_{N_r} = \sum_{r=0}^3 \epsilon_r \{\log \det(I + X_{N_r}) + \log \det(I - Y_{N_r})\}. \quad (\text{JSA24})$$

This verifies the four signs against the original GCR and GSR definitions, including the minus sign between this return and $\mathfrak{T}$.

There is also an exact morphism to the incoming conditional evaluation covariance. The covariance of Z_M restricted to $\ker J_M$ is L_M : subtract from H_M^{-1} its minimum-section part $H_M^{-1} J_M^* K_M^{-1} J_M H_M^{-1}$ and multiply by Z_M and Z_M^* . The remaining operator is the inverse metric on the embedded subspace $\ker J_M$. Identifying that subspace by multiplication by χ therefore proves

$$L_M = \text{diag}(\chi(\alpha_j)) A_{M-q}^{\chi\sigma} \text{diag}(\overline{\chi(\alpha_j)}). \quad (\text{JSA25})$$

Here $A_n^{\chi\sigma}$ uses the exact Gram of $|\chi(c+iy)|^2 d\sigma$ on $\mathcal{P}_n$. The factors $\chi(\alpha_j)$ are nonzero by coprimality. No cross covariance has been removed in identifying the restricted metric.

14 A narrower finite certificate from the two exact low endpoints

At the two low endpoints JSA21 gives, exactly,

$$\mathfrak{R}_{q-1} = 1, \quad \mathfrak{R}_q = \mathfrak{r}_\chi := \frac{\int |f(c+iy)\chi(c+iy)|^2 d\sigma(y)}{\int |\chi(c+iy)|^2 d\sigma(y)} > 0. \quad (\text{JSA26})$$

These are finite original polynomial moments, whose values can also be calculated by expanding in the p_n basis and JSA6. In particular no low-endpoint logarithmic series is needed. Their exact sum is

$$E_{\text{low}} := \log \mathfrak{d}_{q-1} + \log \mathfrak{d}_q = 4 \log |\det M_f| - \log D_{l,q-1} - \log D_{l,q} + \log \mathfrak{r}_\chi. \quad (\text{JSA27})$$

For direct comparison with the incoming certificate retain its same common scalar $s = 2 + \sum_{r=0}^3 \text{Tr } X_{N_r}$ and put $Z_N = I - (I + X_N)/s$. Each Z_N is positive self-adjoint in Ω_N and has spectrum in $[0, 1)$: every eigenvalue of X_N is nonnegative and no larger than $\text{Tr } X_N \leq s - 2$. Together with JSA18 this proves the same assertion for Y_N in its own metric. For any such matrix T define

$$b_p(T) = \sum_{n=1}^p \frac{\text{Tr } T^n}{n}, \quad a_p(T) = \text{Tr } T^p, \quad u_p(T) = b_p(T) + \frac{b_{2p}(T) - b_p(T)}{1 - a_p(T)}. \quad (\text{JSA28})$$

There is a finite p with $a_p(T) < 1$, because if $\rho < 1$ is the largest eigenvalue then $a_p(T) \leq l\rho^p \rightarrow 0$. For $0 \leq x < 1$ put $r_p(x) = \sum_{n>p} x^n/n$. Termwise, $r_{2p}(x) \leq x^p r_p(x)$. Sum over the eigenvalues; the largest x^p is at most $a_p(T)$. Writing the total tail as the sum from $p+1$ to $2p$ plus the tail above $2p$ yields

$$b_p(T) \leq -\log \det(I - T) \leq u_p(T), \quad (\text{JSA29})$$

by moving a_p times the tail to the left. Both bounds converge to the exact value as $p \rightarrow \infty$ along admitted orders.

For $r = 2, 3$ choose admitted orders independently on Z_{N_r} and Y_{N_r} and set

$$L_r = b_{p_r}(Z_{N_r}) + b_{q_r}(Y_{N_r}), \quad U_r = u_{p_r}(Z_{N_r}) + u_{q_r}(Y_{N_r}).$$

The complete return has the following finite enclosure:

$$\boxed{E_{\text{low}} - 2l \log s + L_2 + L_3 \leq R_k^0 - R_k^\sigma \leq E_{\text{low}} - 2l \log s + U_2 + U_3.} \quad (\text{JSA30})$$

To prove it, use $\log \mathfrak{d}_N = l \log s + \log \det(I - Z_N) + \log \det(I - Y_N)$ at the two high endpoints in JSA24 and substitute JSA27 at the two low endpoints. JSA29 then proves both bounds. This uses only four trace series, each with its full specified original metric and mixed factor.

The comparison with the incoming eight-series certificate is exact. If the incoming low-endpoint bounds are $[L_0, U_0]$ and $[L_1, U_1]$ for the corresponding positive log tails, their true sum is $2l \log s - E_{\text{low}}$. It belongs to $[L_0 + L_1, U_0 + U_1]$. Therefore both endpoints in JSA30 are inside the incoming interval $[L_2 + L_3 - U_0 - U_1, U_2 + U_3 - L_0 - L_1]$. Its width decreases by exactly $(U_0 - L_0) + (U_1 - L_1)$

when the same high-endpoint orders and exact low moments are used. If numerical moment values are used, their proved errors must be added to JSA30; the zero-width statement refers to the exact quantities JSA26.

For a direct combination with the existing relation estimates, write

$$C_{\text{rel}} = \log D_{l,q-1} + \log D_{l,q} - \log \mathfrak{r}_\chi - \log D_{l,2q-1} - \log D_{l,2q}.$$

JSA22 and JSA24 prove the exact equality

$$R_k^0 - R_k^\sigma = -C_{\text{rel}} - \log \mathfrak{R}_{2q-1} - \log \mathfrak{R}_{2q}. \quad (\text{JSA31})$$

Thus the proved GSR bounds $\ell_N \leq \log \mathfrak{R}_N \leq u_N$ on these very same relation determinants give the interval $[-C_{\text{rel}} - u_{2q-1} - u_{2q}, -C_{\text{rel}} - \ell_{2q-1} - \ell_{2q}]$. Intersecting this interval with JSA30 is justified by JSA31, not by treating the two constructions as independent models. All surviving original-source error terms still enter the arithmetic interval with their actual proved original-degree estimates. A fixed-input convergent certificate alone makes no tensor-uniform assertion.

15 The reduced certificate also has exact rational bounds

The rational construction RGC1–11 is valid for the full original parameter domain. Its exact factorization is, with $d = \delta^2$ and $g = \gamma^2$,

$$\chi(c+x) = \prod_{a,b \in \{1,3,\dots,k\}} ((x^2 + a^2d + b^2g)^2 - 4a^2dx^2)^e. \quad (\text{JSA32})$$

Expanding the four original factors with roots $a\delta + ib\gamma$, $a\delta - ib\gamma$, $-a\delta + ib\gamma$, $-a\delta - ib\gamma$ gives the displayed factor. Every quartet is included once, so its degree is $4((k+1)/2)^2e = q$ and its coefficients lie in $\mathbb{Q}[d,g]$. Translation by c is the exact triangular binomial map with inverse translation by $-c$; both determinants are one.

In these coordinates $p_n(c+x) \in \mathbb{Q}[x]$, and the coefficient columns of their remainders modulo JSA32 lie in $\mathbb{Q}[d,g]^q$. Their evaluation columns at $x = t_j$ lie in $\mathbb{Q}^l$. Write $h_n = \omega_n/c_\sigma = n!(1/2)_n$. All covariance sums are c_σ^{-1} times rational matrices:

$$\begin{aligned} \hat{K}_N &= \sum_{n=0}^N v_n v_n^\top / h_n, & \hat{C}_M &= \sum_{n=0}^M v_n w_n^\top / h_n, & \hat{A}_M &= \sum_{n=0}^M w_n w_n^\top / h_n, \\ K_N &= c_\sigma^{-1} \hat{K}_N, & C_M &= c_\sigma^{-1} \hat{C}_M, & A_M &= c_\sigma^{-1} \hat{A}_M. \end{aligned} \quad (\text{JSA33})$$

Here the coefficient vectors are real, so transpose is their actual adjoint. JSA12–13 prove that the inverse denominators never vanish on $0 < d < 1/4$, $g > 4$. If $D_N = \text{diag}(h_{N+1}, \dots, h_{N+l})$, substituting the mass factors of JSA33 and $\Omega_N = c_\sigma D_N$ in JSA16 gives

$$X_N = D_N^{-1} U_N^\top \hat{K}_N^{-1} U_N, \quad Y_N = \hat{A}_M^{-1} \hat{C}_M^\top \hat{K}_M^{-1} \hat{C}_M. \quad (\text{JSA34})$$

These are exact equalities of the original endomorphisms and their retained metrics; in particular their entries and all power traces lie in $\mathbb{Q}(d,g)$.

The low-endpoint constant in JSA30 has the exact expression

$$E_{\text{low}} - 2l \log s = \log H_{\text{low}}, \quad H_{\text{low}} = \frac{|\det M_f|^4 \mathfrak{r}_\chi}{D_{l,q-1} D_{l,q} s^{2l}} \in \mathbb{Q}(d,g)_{>0}. \quad (\text{JSA35})$$

To prove rationality of each factor, $\det M_f$ is the determinant of polynomial multiplication in the rational remainder algebra of JSA32, so it is a real polynomial in d, g . It is nonzero by JSA10. The source Gram for real coefficient polynomials in x is c_σ times a rational matrix by the rational monic triangular basis and JSA6. Consequently the common c_σ cancels in the moment ratio $\mathfrak{r}_\chi$,

giving a positive rational function. The other factors are explicitly rational. Also $H_{\text{low}} < 1$: each Z_{N_r} has strictly positive eigenvalues, since $1 + \lambda_{\max}(X_{N_r}) \leq s - 1 < s$; hence each exact positive tail $-\log \det(I - Z_{N_r})$ is strictly positive. The equality before JSA30 at $r = 0, 1$ gives $\log H_{\text{low}}$ as the negative of their sum together with the two nonnegative Y tails.

Put $z = (1 - H_{\text{low}})/(1 + H_{\text{low}}) \in (0, 1)$ and define

$$b_p^{\text{sc}} = 2 \sum_{j=0}^{p-1} \frac{z^{2j+1}}{2j+1}, \quad e_p^{\text{sc}} = \frac{2z^{2p+1}}{(2p+1)(1-z^2)}. \quad (\text{JSA36})$$

Integrating $2/(1-z^2)$ from zero to z gives $-\log H_{\text{low}} = 2 \sum_{j \geq 0} z^{2j+1}/(2j+1)$. Bounding the denominator of every omitted term below by $2p+1$ and summing its geometric numerator proves

$$-b_p^{\text{sc}} - e_p^{\text{sc}} \leq \log H_{\text{low}} \leq -b_p^{\text{sc}}. \quad (\text{JSA37})$$

All terms are rational functions in d, g , and e_p^{sc} tends to zero. Substituting these two bounds into JSA30 gives the completely rational enclosure

$$\boxed{-b_p^{\text{sc}} - e_p^{\text{sc}} + L_2 + L_3 \leq R_k^0 - R_k^\sigma \leq -b_p^{\text{sc}} + U_2 + U_3.} \quad (\text{JSA38})$$

At rational d, g it is computable with exact rational arithmetic. It uses the four full high-endpoint matrix series and one scalar series. Its width is the exact-low-endpoint width in JSA30 plus the explicit scalar error e_p^{sc} . This error can be made arbitrarily small; no assertion about its numerical speed is needed to prove termination.

Finally fix a closed rectangle strictly inside $(0, 1/4) \times (4, \infty)$. The rational denominators above are nonzero on that rectangle, and all matrices and positive metrics are continuous. Their self-adjoint similarities from JSA18 are continuous, so the largest eigenvalues of Z_N, Y_N have a common upper bound less than one by compactness. The continuous scalar z has the same property. The matrix and scalar tail widths therefore converge uniformly to zero. For a fixed admitted order, rational interval evaluation on successively subdivided rectangles converges uniformly because the finitely many denominators have positive absolute minima. This proves finite certification to any prescribed positive excess over the actual variation of the return on the rectangle. The parameter variation is retained. A calibration at rational d, g asserts no relation between those chosen roots and zeros of 2ξ .

16 Exact coefficient map for the incoming mixed-word receiver

The incoming reduced pair receiver also has an exact dictionary to the existing MWR receiver. Retain the incoming nodal order $(c_\pm - i\gamma, c_\pm + i\gamma)$ and let

$$E_0 = \begin{pmatrix} 1 & -i\gamma \\ 1 & i\gamma \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -\gamma^2 \\ 1 & 0 \end{pmatrix}, \quad z_0 = A + i\gamma B, \quad A, B \in \mathbb{R}.$$

The actual reflected unit values give $U_+^{\text{nod}} = \text{diag}(\bar{z}_0, z_0)$ and $U_-^{\text{nod}} = \text{diag}(z_0, \bar{z}_0)$. Direct multiplication gives the full coefficient identities

$$E_0^{-1} U_+^{\text{nod}} E_0 = AI + BJ, \quad E_0^{-1} U_-^{\text{nod}} E_0 = AI - BJ, \quad \det(AI \pm BJ) = A^2 + \gamma^2 B^2 > 0. \quad (\text{JSA39})$$

These are the reduced matrices of the same nonzero unit; no independent phase for the two legs has been supplied. With the exact incoming translation of the oriented periods, set $M = W^{-1} \Pi_{\text{cen}}$ and $F_\pm = (c_\pm^3/3 + c_\pm \gamma^2)/u$. The two nodal receiver matrices therefore satisfy

$$T_\pm = e^{F_\pm} M (AI \pm BJ) E_0^{-1}, \quad T_- T_+^{-1} = e^{F_- - F_+} M (AI - BJ) (AI + BJ)^{-1} M^{-1}. \quad (\text{JSA40})$$

This proves the incoming exceptional off-diagonal equations are exactly the MWR equations multiplied by the same nonzero scalar. For a word with a plus and b minus factors the full tensor map contains the scalar $e^{aF_+ + bF_-}$ and the exact input map $(E_0^{-1})^{\otimes(a+b)}$. Restriction to the unscaled symmetric orbit columns gives their literal symmetric powers; their inverse comes from $E_0^{\otimes(a+b)}$. This transports the coefficient matrix and its entire kernel, with the stated scalar retained in every metric pullback. It identifies the incoming $k = 5, (a, b) = (2, 3)$ rank calculation with that same specialization of the existing MWR construction rather than introducing a different mixed family.

The free Gamma boundary: adjunction and signed expansion

Scope and provenance. This is an independent proof and continuation of Section 5 of `Tau_Gamma_Joint_Schur_Return/NOTE.tex` in the supplied 14 September 2026 package. Its determinant and finite bounds are correct. The expansion proved below strengthens that section while retaining its actual measure, real evaluation nodes, polynomial degree, and all four endpoints. It evaluates the free boundary contribution. The original relation-dependent Schur complement remains a separate summand in the exact signed return, and is not assigned the value of this contribution.

17 The actual source and its polynomial evaluation covariance

Let $l \geq 1$, $k = 4l + 1$, $c = k/2$, $\vartheta = 1/2$, and

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad c_\sigma = \sqrt{2\pi}, \quad \alpha_j = c + 2j + \frac{1}{2} \quad (0 \leq j < l).$$

The polynomial space $\mathcal{P}_M = \{P \in \mathbb{C}[S] : \deg P \leq M\}$ carries the original inner product

$$\langle P, Q \rangle_\sigma = \int_{\mathbb{R}} P(c + iy) \overline{Q(c + iy)} \sigma(y) dy.$$

The evaluation map is $E_M : \mathcal{P}_M \rightarrow \mathbb{C}^l$, $E_M P = (P(\alpha_j))_{j=0}^{l-1}$. Its covariance $A_M = E_M E_M^*$ is in the stated Euclidean target metric; no probability-mass rescaling is made.

Lemma 17.1 (The original monic system and its mass). *The monic polynomials $p_n \in \mathbb{C}[S]$ obey*

$$p_0 = 1, \quad p_1 = S - c, \quad p_{n+1} = (S - c)p_n + n(n - \tfrac{1}{2})p_{n-1}, \quad \|p_n\|_\sigma^2 = c_\sigma n! (\vartheta)_n.$$

Proof. For $a > 0$, set $f_a(t) = \exp(at - e^t)$. The substitution $r = e^t$ in the Gamma integral gives $\Gamma(a + iy/2) = \int_{\mathbb{R}} f_a(t) e^{iyt/2} dt$. Plancherel with this explicit frequency $y/2$ gives, for real u ,

$$\begin{aligned} \int_{\mathbb{R}} \frac{|\Gamma(a + iy/2)|^2}{2\pi} e^{iuy} dy &= 2 \int_{\mathbb{R}} f_a(t) f_a(t + 2u) dt \\ &= \frac{2\Gamma(2a)e^{2au}}{(1 + e^{2u})^{2a}} = 2^{1-2a} \Gamma(2a) (\cosh u)^{-2a}. \end{aligned}$$

In particular the mass for $a = 1/4$ is $c_\sigma = \sqrt{2\pi}$. Every derivative of f_a belongs to $L^2(\mathbb{R})$, since it is f_a times a polynomial in e^t ; Plancherel therefore gives every required polynomial moment of $|\Gamma(a + iy/2)|^2$.

Define, for real z near zero,

$$F(x, z) = (1 - z^2)^{-1/4} \exp(x \operatorname{arctanh} z) = \sum_{n \geq 0} p_n(c + x) \frac{z^n}{n!}.$$

The identity $(1 - z^2)\partial_z F = (x + z/2)F$ gives the displayed recurrence and monicity. For real z, w near zero the Fourier formula gives

$$\begin{aligned} \int_{\mathbb{R}} F(iy, z) \overline{F(iy, w)} \sigma(y) dy &= (1 - z^2)^{-1/4} (1 - w^2)^{-1/4} c_\sigma (\cosh(\operatorname{arctanh} z - \operatorname{arctanh} w))^{-1/2} \\ &= c_\sigma (1 - zw)^{-1/2}. \end{aligned}$$

Differentiating at $(z, w) = (0, 0)$ is justified by the polynomial moments just established. It gives orthogonality and the stated norms, including the full factor c_σ . $\square$

At $x = 2j + 1/2$, the same generating function is $(1 + z)^j(1 - z)^{-j-1/2}$. Consequently

$$p_n(\alpha_j) = (\vartheta)_n Q_j(n), \quad Q_j(n) = \sum_{r=0}^j \binom{j}{r} \frac{2^r n^r}{(\vartheta)_r}. \quad (85)$$

Indeed summing the right side divided by $n!$ gives

$$\sum_{r=0}^j \binom{j}{r} (2z)^r (1 - z)^{-\vartheta-r} = (1 + z)^j (1 - z)^{-j-\vartheta}.$$

Thus Q_j has degree j and leading coefficient $2^j/(\vartheta)_j$, and the actual covariance is

$$(A_M)_{ij} = c_\sigma^{-1} \sum_{n=0}^M \frac{(\vartheta)_n}{n!} Q_i(n) Q_j(n). \quad (86)$$

18 Adjoint polynomial maps in both original measures

For this section only, allow any $\vartheta > 0$ and any integer $M \geq 0$. Put $w_n = (\vartheta)_n/n!$ and $K_M = (\vartheta)_{M+1}/M!$. The discrete space uses $\sum_{n=0}^M w_n \delta_n$ and the continuous space on $[0, 1]$ uses $K_M t^{\vartheta-1} dt$. The positive joint measure

$$K_M \binom{M}{n} t^{n+\vartheta-1} (1-t)^{M-n} dt \quad (0 \leq n \leq M)$$

has the continuous marginal by the binomial theorem and the discrete marginal because

$$K_M \binom{M}{n} B(n + \vartheta, M - n + 1) = \frac{\Gamma(n + \vartheta)}{\Gamma(\vartheta)n!} = w_n.$$

Its common mass is K_M/ϑ ; neither measure is replaced by a probability measure. It gives the exact adjunction

$$B : \mathbb{C}^{\{0, \dots, M\}} \longrightarrow L^2([0, 1], K_M t^{\vartheta-1} dt), \quad (BP)(t) = \sum_{n=0}^M \binom{M}{n} t^n (1-t)^{M-n} P(n),$$

$$B^*(t^r)(n) = \frac{(n + \vartheta)_r}{(M + \vartheta + 1)_r}, \quad \langle BP, R \rangle_{\text{cont}} = \langle P, B^* R \rangle_{\text{disc}}.$$

The second formula follows from the ratio of the two beta integrals, so it proves both the map and its adjoint in the stated metrics.

Lemma 18.1 (Exact discrete monic norms). *For $0 \leq j \leq M$, the squared norm of the discrete monic orthogonal polynomial of degree j is*

$$\nu_j(M) = \frac{(\vartheta)_{M+j+1} (j!)^2}{(M-j)! (2j + \vartheta) (j + \vartheta)_j^2}. \quad (87)$$

Proof. The continuous monic orthogonal polynomial is

$$P_j(t) = \frac{(-1)^j}{(j + \vartheta)_j} t^{1-\vartheta} \frac{d^j}{dt^j} \{t^{j+\vartheta-1} (1-t)^j\}.$$

Its leading coefficient is one. Integration by parts j times proves orthogonality against every smaller degree. For each derivative of order $s \leq j - 1$ of the bracket, the factor at zero is $O(t^{j+\vartheta-1-s})$,

with exponent at least $\vartheta > 0$; at one the remaining vanishing order is at least $j - s \geq 1$. Every boundary term therefore vanishes. Applying the same integration by parts to its own norm gives

$$\|P_j\|_{\text{cont}}^2 = \frac{K_M j!}{(j + \vartheta)_j} B(j + \vartheta, j + 1) = \frac{K_M (j!)^2}{(2j + \vartheta)(j + \vartheta)_j^2}.$$

Let Q_j^{mon} be the discrete monic polynomial. The exact identity $B(n^j) = M^j t^j$ and the displayed formula for B^* prove preservation of degrees and their leading coefficients. Adjunction against every lower degree then proves

$$BQ_j^{\text{mon}} = M^j P_j, \quad B^* P_j = \frac{Q_j^{\text{mon}}}{(M + \vartheta + 1)_j}.$$

Evaluating $\langle BQ_j^{\text{mon}}, P_j \rangle$ on the two sides of the adjunction yields $\nu_j(M) = M^j (M + \vartheta + 1)_j \|P_j\|_{\text{cont}}^2$. Substitution gives (87), also for $j = 0$. $\square$

Return to $\vartheta = 1/2$. The triangular change from n^j to $Q_j(n)$ in (86) has determinant $\prod_{j < l} 2^j / (\vartheta)_j$; the monic orthogonal change has determinant one. For every $M \geq l - 1$,

$$\det A_M = c_\sigma^{-l} \prod_{j=0}^{l-1} \frac{4^j (j!)^2 (\vartheta)_{M+j+1}}{(\vartheta)_j^2 (M-j)! (2j + \vartheta)(j + \vartheta)_j^2}. \quad (88)$$

In particular all determinants are strictly positive, and

$$\frac{\det A_{M+1}}{\det A_M} = \prod_{j=0}^{l-1} \frac{M + j + \vartheta + 1}{M + 1 - j}. \quad (89)$$

19 All four endpoints and a sharper finite expansion

For every integer $q \geq 1$ put

$$C_{q,l} = \log \frac{\det A_{2q+l-1} \det A_{2q+l}}{\det A_{q+l-1} \det A_{q+l}}, \quad B_l = l(l - \tfrac{1}{2}).$$

The constants in (88) are retained until this exact signed combination is taken. Set $a_j = l - j$, $b_j = l + j + 1/2$, and $f_j(x) = \log((x + b_j)/(x + a_j))$. Formula (89) gives

$$C_{q,l} = \sum_{j=0}^{l-1} \sum_{n=q}^{2q-1} \{f_j(n) + f_j(n+1)\}. \quad (90)$$

All summands are positive. Applying $u/(1+u) \leq \log(1+u) \leq u$ to these exact two windows, whose upper denominators are at least $n+1$ and lower denominators at most $n+2l+1/2$, gives

$$2B_l \log \frac{2q+2l+1/2}{q+2l+1/2} \leq C_{q,l} \leq 2B_l \log 2.$$

Here the decreasing-integrand comparisons are respectively $\sum_{n=q}^{2q-1} (n+d)^{-1} \geq \int_q^{2q} (x+d)^{-1} dx$ and $\sum_{n=q}^{2q-1} (n+1)^{-1} \leq \int_q^{2q} x^{-1} dx$. These justify the incoming finite bounds without an asymptotic step.

Theorem 19.1 (Uniform finite signed expansion). *For every integer $q, l \geq 1$,*

$$C_{q,l} = 2B_l \log 2 - \frac{B_l(l+1/4)}{q} + R_{q,l}, \quad 0 \leq R_{q,l} \leq \frac{B_l(14l^2 + 5l + 3)}{16q^2}. \quad (91)$$

There is also the exact second-order enclosure

$$C_{q,l} = 2B_l \log 2 - \frac{B_l(l+1/4)}{q} + \frac{l^2(2l-1)(14l+5)}{32q^2} + E_{q,l}, \quad (92)$$

$$-\frac{7l(2l-1)(4l+1)(12l^2+2l-1)}{768q^3} \leq E_{q,l} \leq \frac{3B_l}{16q^2}. \quad (93)$$

Proof. For a twice differentiable f , integration by parts on one unit interval gives the exact trapezoidal identity

$$\frac{f(n) + f(n+1)}{2} - \int_n^{n+1} f(x) dx = \frac{1}{2} \int_0^1 s(1-s) f''(n+s) ds.$$

For our function $f_j(x) = \int_{a_j}^{b_j} (x+t)^{-1} dt$, its second derivative is positive. Since $s(1-s)/2 \leq 1/8$, summing the identity and using (90) gives

$$\begin{aligned} C_{q,l} &= 2 \sum_j \int_q^{2q} f_j(x) dx + T_{q,l}, \\ 0 \leq T_{q,l} &\leq \frac{1}{4} \sum_j \{f'_j(2q) - f'_j(q)\} \\ &= \frac{1}{4} \sum_j \int_{a_j}^{b_j} \left(\frac{1}{(q+t)^2} - \frac{1}{(2q+t)^2} \right) dt \leq \frac{3B_l}{16q^2}. \end{aligned}$$

The final integrand decreases for $t \geq 0$ and at zero equals $3/(4q^2)$; also $\sum_j (b_j - a_j) = B_l$. Fubini, in positive finite integrals, gives

$$2 \sum_j \int_q^{2q} f_j(x) dx = 2 \sum_j \int_{a_j}^{b_j} g_q(t) dt, \quad g_q(t) = \log \frac{2q+t}{q+t}.$$

For $t \geq 0$, direct differentiation shows

$$g_q(0) = \log 2, \quad g'_q(0) = -\frac{1}{2q}, \quad 0 < g''_q(t) \leq \frac{3}{4q^2}, \quad -\frac{7}{4q^3} \leq g'''_q(t) < 0.$$

Taylor's formula with integral remainder therefore gives both

$$\begin{aligned} 0 \leq g_q(t) - \log 2 + \frac{t}{2q} &\leq \frac{3t^2}{8q^2}, \\ -\frac{7t^3}{24q^3} \leq g_q(t) - \log 2 + \frac{t}{2q} - \frac{3t^2}{8q^2} &\leq 0. \end{aligned}$$

The finite power sums, obtained by expansion and $\sum_{j < l} j = l(l-1)/2$, $\sum_{j < l} j^2 = l(l-1)(2l-1)/6$, $\sum_{j < l} j^3 = l^2(l-1)^2/4$, are

$$\begin{aligned} \sum_j \int_{a_j}^{b_j} t dt &= B_l(l + \tfrac{1}{4}), \\ \sum_j (b_j^3 - a_j^3) &= \frac{l^2(2l-1)(14l+5)}{8}, \\ \sum_j (b_j^4 - a_j^4) &= \frac{l(2l-1)(4l+1)(12l^2+2l-1)}{16}. \end{aligned}$$

Integration of the first Taylor enclosure, together with the bound on $T_{q,l}$, gives (91), because

$$\frac{1}{4} \sum_j (b_j^3 - a_j^3) + \frac{3B_l}{16} = \frac{B_l(14l^2+5l+3)}{16}.$$

Integration of the second enclosure gives (92) and (93); its lower bound is $-7 \sum_j (b_j^4 - a_j^4)/(48q^3)$ and its upper bound is the retained positive trapezoidal error. This proves both finite formulas. $\square$

20 The stipulated packet degree is retained in the limits

The programme uses precisely

$$q = (4l + 2)^2(1 + (4l + 1)(m - 1)), \quad m \geq 1.$$

For $m = 1$, substitution into (92) gives

$$C_{q,l} = 2l^2 \log 2 - l(\log 2 + \frac{1}{16}) + \frac{167}{2048} + O(l^{-1}). \quad (94)$$

Every error in this statement is bounded by (93) and a rational remainder from the displayed value of q . In detail,

$$\frac{B_l(l + 1/4)}{(4l + 2)^2} = \frac{l}{16} - \frac{5}{64} + O(l^{-1}), \quad \frac{l^2(2l - 1)(14l + 5)}{32(4l + 2)^4} = \frac{7}{2048} + O(l^{-1}),$$

and (93) is $O(l^{-1})$. Hence $C_{q,l}/q \rightarrow (\log 2)/8$ and the constant term in (94) is $5/64 + 7/2048 = 167/2048$.

For each fixed integer $m > 1$, $q/l^3 \rightarrow 64(m - 1)$. Formula (91) has $R_{q,l} = O(l^{-2})$, while $B_l(l + 1/4)/q \rightarrow 1/(64(m - 1))$. Thus

$$C_{q,l} - 2l(l - \frac{1}{2}) \log 2 \longrightarrow -\frac{1}{64(m - 1)}, \quad \frac{lC_{q,l}}{q} \longrightarrow \frac{\log 2}{32(m - 1)}. \quad (95)$$

The relation-dependent matrices at the same four endpoints remain in the exact total return. Equations (94)–(95) evaluate its explicitly identified free summand, with a uniform finite error already proved above; they impose no value on that remaining summand.

Exact rational construction of the Gamma certificate

21 Exact rational construction of the joint Gamma certificate

Fix integers $l \geq 1, m \geq 1$, and retain

$$k = 4l + 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = (k + 1)^2 e, \quad d = \delta^2 \in (0, 1/4), \quad g = \gamma^2 > 4.$$

The original coefficient is S , and $x = S - c$. Translation is the explicit invertible coefficient map

$$S^n = \sum_{j=0}^n \binom{n}{j} c^{n-j} x^j, \quad x^n = \sum_{j=0}^n \binom{n}{j} (-c)^{n-j} S^j. \quad (\text{RGC1})$$

Both triangular matrices have determinant one. Every polynomial, remainder and metric below is carried through these matrices; the original source norm is retained.

21.1 The complete packet and all polynomial coefficients

In the translated variable the original packet is exactly

$$\chi(c + x) = \prod_{a,b \in \{1,3,\dots,k\}} [(x^2 + a^2 d + b^2 g)^2 - 4a^2 d x^2]^e. \quad (\text{RGC2})$$

Indeed the factor displayed is the product of $(x - a\delta - ib\gamma)$, $(x - a\delta + ib\gamma)$, $(x + a\delta - ib\gamma)$ and $(x + a\delta + ib\gamma)$. The positive odd integers index each quartet exactly once because k is odd. Its degree is $4((k + 1)/2)^2 e = q$, with all original multiplicities present. Consequently every coefficient belongs to $\mathbb{Q}[d, g]$.

Use the original monic sigma polynomials, written in the translated coordinate:

$$p_0 = 1, \quad p_1 = x, \quad p_{n+1} = x p_n + n(n - \tfrac{1}{2}) p_{n-1}, \quad h_n = n! (\tfrac{1}{2})_n, \quad \omega_n^\sigma = c_\sigma h_n, \quad c_\sigma = \sqrt{2\pi}. \quad (\text{RGC3})$$

Thus $p_n \in \mathbb{Q}[x]$, $h_n \in \mathbb{Q}_{>0}$. Let $v_n \in \mathbb{Q}[d, g]^q$ be the coefficient column of $\text{rem}_\chi p_n$ in $1, x, \dots, x^{q-1}$. These entries are polynomial because division by the monic polynomial RGC2 uses no variable denominator. Let

$$w_n = (p_n(2j + \tfrac{1}{2}))_{j=0}^{l-1} \in \mathbb{Q}^l. \quad (\text{RGC4})$$

This is evaluation at the original nodes $\alpha_j = c + 2j + 1/2$, after the exact translation RGC1.

21.2 Mass-preserving covariance identities

For $M \geq q + l - 1$ and $N \geq q - 1$, define the following rational matrices, retaining the original source covariance by its explicit scalar factor:

$$\hat{K}_N = \sum_{n=0}^N v_n v_n^\top / h_n, \quad K_N = c_\sigma^{-1} \hat{K}_N, \quad (96)$$

$$\hat{C}_M = \sum_{n=0}^M v_n w_n^\top / h_n, \quad C_M = c_\sigma^{-1} \hat{C}_M, \quad (97)$$

$$\hat{A}_M = \sum_{n=0}^M w_n w_n^\top / h_n, \quad A_M = c_\sigma^{-1} \hat{A}_M. \quad (\text{RGC5})$$

These equalities follow by expanding the actual inverse source Gram in its complete orthogonal basis $p_0, \dots, p_M$, whose norms are exactly $c_\sigma h_n$. All vectors are real for real d, g ; transpose is therefore the actual Hermitian adjoint. In particular

$$G_N^\sigma = c_\sigma \widehat{K}_N^{-1}. \quad (\text{RGC6})$$

The covariance $\widehat{K}_N$ is positive definite: its first q columns are the coefficient vectors of $p_0, \dots, p_{q-1}$, a triangular basis with diagonal one. The joint covariance of (v_n, w_n) is positive definite for $M \geq q + l - 1$. To prove this assertion with the original maps, the remainder map to $\mathbb{R}[x]/\chi(c+x)$ and the evaluation map at the l distinct real points $2j+1/2$ are jointly onto by monic division and interpolation. Every root of RGC2 has nonzero imaginary part, so those evaluations are coprime to χ . Their joint kernel in $\mathcal{P}_M$ is precisely $\chi f \mathcal{P}_{M-q-l}$. Applying this onto map to the positive source Gram proves the assertion and both strict Schur inequalities. No determinant denominator used below vanishes on the stated parameter domain.

For $M = N + l$ put $U_N = [v_{N+1}, \dots, v_{N+l}]$ and $D_N = \text{diag}(h_{N+1}, \dots, h_{N+l})$. The original metric of the first small endomorphism is $\Omega_N = c_\sigma D_N$, and that of the second is $A_M = c_\sigma^{-1} \widehat{A}_M$. Substitution into the original joint-Schur formulas gives the exact equalities

$$X_N = D_N^{-1} U_N^\top \widehat{K}_N^{-1} U_N, \quad (98)$$

$$Y_N = \widehat{A}_M^{-1} \widehat{C}_M^\top \widehat{K}_M^{-1} \widehat{C}_M. \quad (\text{RGC7})$$

For example the three scalar factors in $C_M^* G_M^\sigma C_M$ are $c_\sigma^{-1} c_\sigma c_\sigma^{-1} = c_\sigma^{-1}$; the inverse of A_M supplies c_σ . Thus RGC7 is an equality of the original endomorphisms after the proved coordinate map, with their original forms still as specified. Their entries belong to $\mathbb{Q}(d, g)$. Self-adjointness is in these positive forms; ordinary Euclidean symmetry of RGC7 is unnecessary. The actual spectrum of X_N is nonnegative, and that of Y_N lies in $[0, 1)$ by the strict joint Schur inequality already proved.

21.3 A finite rational enclosure of the exact return

Use the original endpoints

$$(N_0, N_1, N_2, N_3) = (q-1, q, 2q-1, 2q), \quad (\varepsilon_0, \varepsilon_1, \varepsilon_2, \varepsilon_3) = (1, 1, -1, -1).$$

Set

$$s = 2 + \sum_{r=0}^3 \text{Tr } X_{N_r}, \quad Z_N = I - (I + X_N)/s. \quad (\text{RGC8})$$

All entries and all power traces of these eight endomorphisms Z_{N_r}, Y_{N_r} are rational functions in d, g . Since $\lambda_{\max}(X_N) \leq \text{Tr } X_N < s-1$, the eigenvalues of Z_N also lie in $[0, 1)$. For any such matrix T , and integer $p \geq 1$, set

$$b_p(T) = \sum_{n=1}^p \frac{\text{Tr } T^n}{n}, \quad a_p(T) = \text{Tr } T^p, \quad u_p(T) = b_p(T) + \frac{b_{2p}(T) - b_p(T)}{1 - a_p(T)}. \quad (\text{RGC9})$$

The last expression is used at an order with $a_p(T) < 1$. Here is a direct proof of its bound. If the eigenvalues are $t_i \in [0, 1)$ and $r_p(t) = \sum_{n>p} t^n/n$, then $r_{2p}(t) \leq t^p r_p(t)$. Summing and using nonnegative summands gives

$$\sum_i r_{2p}(t_i) \leq a_p(T) \sum_i r_p(t_i).$$

The identity $\sum_i r_p(t_i) = b_{2p}(T) - b_p(T) + \sum_i r_{2p}(t_i)$ now proves

$$b_p(T) \leq -\log \det(I - T) \leq u_p(T). \quad (\text{RGC10})$$

Because $a_p(T) \rightarrow 0$, a finite admitted order exists. Both bounds converge to the same logarithm as $p \rightarrow \infty$.

For independently chosen admitted orders in each of the eight blocks, let L_r be the sum of its two lower bounds and U_r the sum of its two upper bounds. The exact four-endpoint identity then proves

$$L_2 + L_3 - U_0 - U_1 \leq R_k^0 - R_k^\sigma \leq U_2 + U_3 - L_0 - L_1. \quad (\text{RGC11})$$

Every finite bound in RGC11 is a rational function evaluated at the actual d, g . In particular for rational d, g the entire certificate can be computed in exact rational arithmetic. The source factor c_σ , the tensor mass C_k , and $\det M_f$ disappeared only through RGC5–7 and the already proved signed cancellation of their four original contributions. No approximate Gamma values or approximate period matrices are inputs to this Gamma-reference certificate.

This also proves a terminating enclosure procedure on every fixed closed rational rectangle strictly inside $(0, 1/4) \times (4, \infty)$. Indeed the entries in RGC7–8 are continuous there and every denominator is nonzero. Strict Schur positivity and compactness imply a common spectral radius $\rho < 1$ for the eight continuous families. Hence their $a_p \leq l\rho^p$ are uniformly less than one at a sufficiently large finite order, and the bound widths converge uniformly to zero. Rational interval arithmetic on subdivided rectangles certifies these inequalities in finitely many subdivisions: polynomial interval bounds converge to point values as the diameters decrease; compactness gives a positive lower bound for the finitely many denominators and for $1 - a_p$. The same argument bounds each rational endpoint value to any prescribed positive interval-arithmetic error. Increasing p and subdividing supplies any prescribed positive excess width over the actual variation of the return on that rectangle. One must retain this variation for a parameter rectangle; an interval cannot in general shrink to a single value until the parameter region shrinks as well.

These claims concern the exact original Gamma comparison on the full packet parameter domain. Evaluating a rational pair (d, g) is a calibration of that family, and makes no claim that its roots are zeros of 2ξ . The original theta norm, its arithmetic defect, its full-unit map and its mixed support are returned by the separate exact receiving equations. The rational construction supplies their specified Gamma term.

21.4 Exact signed determinant and a geometric scalar certificate

There is a faster finite evaluation route at rational d, g that also retains cancellation before any logarithm is bounded. Form the positive rational function

$$\mathcal{D}_k(d, g) = \prod_{r=0}^3 \{\det(I + X_{N_r}) \det(I - Y_{N_r})\}^{\varepsilon_r}. \quad (\text{RGC12})$$

Positivity and RGC7 prove that this lies in $\mathbb{Q}(d, g)_{>0}$, and the exact return is $R_k^0 - R_k^\sigma = \log \mathcal{D}_k(d, g)$ by the four original signs. In particular no separate endpoint-log error need be added when this signed product has first been computed exactly.

For any positive rational r , there is a unique integer h such that $r = 2^h s$, $1 \leq s < 2$. It is obtained by comparing the positive numerator and denominator of r with integer powers of two. This is an exact finite comparison of integers. Put $t = (s - 1)/(s + 1) \in [0, 1/3]$, and for integers $p \geq 0$ set

$$B_p(t) = 2 \sum_{j=0}^{p-1} \frac{t^{2j+1}}{2j+1}, \quad E_p(t) = \frac{2t^{2p+1}}{(2p+1)(1-t^2)}. \quad (\text{RGC13})$$

The sum at $p = 0$ is zero. The identity $\log s = 2 \int_0^t (1 - z^2)^{-1} dz$ and the geometric series give

$$B_p(t) \leq \log s \leq B_p(t) + E_p(t). \quad (\text{RGC14})$$

The proof is termwise integration of the nonnegative series; every omitted denominator is at least $2p + 1$, so its geometric numerator sum is bounded by $E_p(t)$. The same formula at $t = 1/3$ bounds

$\log 2$. Define $b = B_p(t)$, $e = E_p(t)$, $b_2 = B_p(1/3)$ and $e_2 = E_p(1/3)$. Restoring the exact integer exponent gives the entirely rational interval

$$\begin{cases} hb_2 + b \leq \log r \leq h(b_2 + e_2) + b + e, & h \geq 0, \\ h(b_2 + e_2) + b \leq \log r \leq hb_2 + b + e, & h < 0. \end{cases} \quad (\text{RGC15})$$

Its width is $|h|e_2 + e$. Since $0 \leq t \leq 1/3$,

$$0 \leq |h|e_2 + e \leq \frac{3(|h| + 1)}{4(2p + 1)}9^{-p} \leq \frac{3(|h| + 1)}{4}9^{-p}. \quad (\text{RGC16})$$

Every quantity in RGC13–16 is rational. Given any positive rational tolerance ϵ , the least non-negative integer p satisfying $9^p \geq 3(|h| + 1)/(4\epsilon)$ is a finite stopping choice with width at most ϵ . It can be found by exact integer multiplication and comparison. Applying this procedure to $r = \mathcal{D}_k(d, g)$ proves a geometric rational enclosure for the complete signed return. Small positive Schur gaps remain in the exact rational product; they do not make the scalar expansion parameter approach one. The exponent h and all rational numerator/denominator sizes are retained in the construction. This proof asserts no polynomial bit complexity for forming the original large quotient matrices.

For the full degree-36 calibration $k = 5, m = 1, d = 1/16, g = 9$, all four cutoffs 35, 36, 71, 72 were evaluated in exact rational arithmetic, including the complete original relation-Gram dimensions 0, 1, 36, 37. The independent identity checked at each endpoint is

$$\frac{D_{1,N}}{\mathfrak{N}_N}(1 + X_N)(1 - Y_N) = \chi(c + 1/2)^2, \quad D_{1,N} = \prod_{n=0}^N (n + 1/2)(n + 3/2).$$

The fixed multiplication determinant on the right cancels with the four signs. Exact rational data, the full coefficient algorithm and the finite logarithm proof are supplied in `exact_rational/`. This calibration confirms the opposite signs of the incoming return and the GSR functional, on the full stipulated polynomial degrees; it asserts no zeta-zero coordinates.

The signed return to the original arithmetic source

22 The original objects and both coordinate maps

Fix the original complete quartet with $0 < \delta < 1/2$, $\gamma > 2$, multiplicity $m \geq 1$, $k = 4l + 1$, $l \geq 1$, and retain

$$\begin{aligned} c &= k/2, \quad e_k = 1 + k(m-1), \quad q = (k+1)^2 e_k, \\ \lambda_{ab} &= c + (2a-k)\delta + i(2b-k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^{e_k}, \\ t_j &= 2j + \frac{1}{2}, \quad \alpha_j = c + t_j, \quad f(S) = \prod_{j=0}^{l-1} (S - \alpha_j), \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \tag{JSR1}$$

All remainders use the increasing coefficient frame $1, S, \dots, S^{q-1}$. Since k is odd, $\text{Im } \lambda_{ab} = (2b-k)\gamma \neq 0$, whereas each α_j is real. Thus f and χ are coprime. The original multiplication $M_f : E \rightarrow E$ is invertible, with inverse its Bezout remainder. No original Taylor unit, ordered tensor inclusion, source map or boundary observation is altered.

The two original reference measures and their masses are

$$\begin{aligned} d\sigma(y) &= \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \int d\sigma = \sqrt{2\pi}, \quad S = c + iy, \\ dm_{0,k} &= \beta_k |f(c + iy)|^2 d\sigma, \quad \int dm_{0,k} = (2\pi)^{k/2}, \\ \beta_k &= \frac{(2\pi)^{k/2}}{\sqrt{2} \Gamma(k/2)} = \frac{(\sqrt{2\pi})^k}{c_{k/4}} 4^{-l}, \quad c_a = 2^{1-2a} \Gamma(2a). \end{aligned} \tag{JSR2}$$

The last equality follows because $4^l c_{k/4} = 2^{2l+1-k/2} \Gamma(k/2) = \sqrt{2} \Gamma(k/2)$. Consequently the new note's C_k is exactly GCR's β_k . Gamma recurrence gives the measure identity with the original coordinate and both masses retained.

For $J_N : \mathcal{P}_N \rightarrow E$ the original remainder map, let $\mathbf{E} : E \rightarrow \mathbb{C}^q$ be the divided Hermite jet map, with rows (a, b, r) in the original order:

$$(\mathbf{E})_{(a,b,r),n} = \binom{n}{r} \lambda_{ab}^{n-r}, \quad \mathbf{D} = \text{diag}_{a,b,0 \leq r < e_k} (r!). \tag{JSR3}$$

Entries with $n < r$ are zero. The raw jet map in the new note is exactly $\mathbf{E}^{\text{raw}} = \mathbf{D}\mathbf{E}$, and hence its inverse is $\mathbf{E}^{-1}\mathbf{D}^{-1}$. The confluent Vandermonde determinant of $\mathbf{E}$ is $\prod_{\lambda < \mu} (\mu - \lambda)^{e_k^2}$; the raw map retains in addition $\det \mathbf{D} = \prod_{a,b,r} r!$. On the complete primary jets let $\mathbf{U}_f$ be multiplication by f , with entries $f^{[r-s]}(\lambda)$ for $r \geq s$. Then

$$\mathbf{E}M_f = \mathbf{U}_f\mathbf{E}, \quad \mathbf{F}_\chi^{\text{raw}} = \mathbf{D}\mathbf{U}_f\mathbf{D}^{-1}, \quad \det M_f = \det \mathbf{U}_f = \prod_{a,b} f(\lambda_{ab})^{e_k}. \tag{JSR4}$$

Indeed the (r, s) entry after this conjugation is $r!f^{(r-s)}(\lambda)/(s!(r-s)!) = \binom{r}{s}f^{(r-s)}(\lambda)$, exactly the raw Leibniz matrix. This proves the coordinate dictionary on every nilpotent direction, including all derivative factorials.

23 The two metrized small matrices and the full mixed determinant

For $N \geq q - 1$ put $M = N + l$. Write all covariances in the original coefficient remainder frame. With $Z_M P = (P(\alpha_j))_{0 \leq j < l}$ and the full sigma source Gram H_M^σ , define

$$\begin{pmatrix} J_M \\ Z_M \end{pmatrix} (H_M^\sigma)^{-1} \begin{pmatrix} J_M^* & Z_M^* \end{pmatrix} = \begin{pmatrix} K_M & C_M \\ C_M^* & A_M \end{pmatrix}, \quad G_M^\sigma = K_M^{-1}. \quad (\text{JSR5})$$

The joint observation is onto: division by the coprime monic polynomial $f\chi$ realizes every joint remainder at degree $q + l - 1 \leq M$. Its kernel is $f\chi\mathcal{P}_{N-q}$. Thus the joint covariance is positive definite and both $S_M = K_M - C_M A_M^{-1} C_M^*$ and $L_M = A_M - C_M^* K_M^{-1} C_M$ are positive definite. Multiplication $P \mapsto fP$ maps the original fibre $J_N P = v$ bijectively onto the joint fibre $(M_f v, 0)$. Its inverse is polynomial division on $\ker Z_M$. Minimization in this exact fibre and JSR2 give

$$G_N^0 = \beta_k M_f^* S_M^{-1} M_f. \quad (\text{JSR6})$$

This proof retains the actual cross covariance C_M .

Let p_n be the original monic sigma polynomials in S . Their norms are exactly $\gamma_n = \sqrt{2\pi}(2n)!/4^n = \sqrt{2\pi} n!(1/2)_n$. In the new note's notation put

$$\begin{aligned} U_N &= [Jp_{N+1}, \dots, Jp_{N+l}], \quad \Omega_N = \text{diag}(\gamma_{N+1}, \dots, \gamma_{N+l}), \\ X_N &= \Omega_N^{-1} U_N^* G_N^\sigma U_N, \quad Y_N = A_M^{-1} C_M^* G_M^\sigma C_M, \\ \hat{X}_N &= \Omega_N^{1/2} X_N \Omega_N^{-1/2}, \quad \hat{Y}_N = A_M^{1/2} Y_N A_M^{-1/2}. \end{aligned} \quad (\text{JSR7})$$

The maps $x \mapsto \Omega_N^{1/2} x$ and $y \mapsto A_M^{1/2} y$ are isometries from their displayed positive metrics to the standard coordinate metric. Thus $\hat{X}_N = \Omega_N^{-1/2} U_N^* G_N^\sigma U_N \Omega_N^{-1/2} \succeq 0$ and $\hat{Y}_N = A_M^{-1/2} C_M^* G_M^\sigma C_M A_M^{-1/2} \succeq 0$. Moreover $I - \hat{Y}_N = A_M^{-1/2} L_M A_M^{-1/2} > 0$. In particular X_N is self-adjoint in Ω_N , and Y_N is self-adjoint in A_M with eigenvalues in $[0, 1)$.

Use tildes to denote precisely the GCR divided-jet matrices, so that no coefficient covariance is identified with a jet covariance without its map. Direct substitution of JSR3 gives

$$\begin{aligned} \tilde{K}_N &= \mathbf{E} K_N \mathbf{E}^*, \quad \tilde{U}_N = \mathbf{E} U_N \Omega_N^{-1/2}, \quad \tilde{V}_N = \mathbf{E} C_M A_M^{-1/2}, \\ \tilde{W}_N &= (\tilde{U}_N \tilde{V}_N), \quad J_l = \text{diag}(I_l, -I_l), \quad \tilde{S}_N = \mathbf{E} S_M \mathbf{E}^*. \end{aligned} \quad (\text{JSR8})$$

The real-node observation has order zero, so no factorial coordinate change occurs on its l -dimensional target. All cross kernels transform by the left factor $\mathbf{E}$. The original finite source kernel sum gives $K_M = K_N + U_N \Omega_N^{-1} U_N^*$; hence $\tilde{S}_N = \tilde{K}_N + \tilde{U}_N \tilde{U}_N^* - \tilde{V}_N \tilde{V}_N^*$, exactly GCR22. The GCR24 positive factors are therefore precisely

$$\begin{aligned} \mathcal{A}_N^{\text{GCR}} &= I + \hat{X}_N = \Omega_N^{1/2} (I + X_N) \Omega_N^{-1/2}, \\ \mathcal{B}_N^{\text{GCR}} &= I - \hat{Y}_N = A_M^{1/2} (I - Y_N) A_M^{-1/2}, \\ \mathfrak{d}_N &:= \frac{\det \tilde{S}_N}{\det \tilde{K}_N} = \det(I_{2l} + J_l \tilde{W}_N^* \tilde{K}_N^{-1} \tilde{W}_N) \\ &= \det(I + X_N) \det(I - Y_N) > 0. \end{aligned} \quad (\text{JSR9})$$

For completeness, the determinant of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ computed by either diagonal block proves $\det(I + AB) = \det(I + BA)$, giving the $2l$ determinant. Alternatively factor $\det S_M / \det K_N$ as $(\det K_M / \det K_N)(\det S_M / \det K_M)$. The two factors are respectively $\det(I + X_N)$ and $\det(I - Y_N)$ by the same identity and the full joint block determinant. This proves JSR9 without requiring the

two differently metrized endomorphisms to commute. In the GCR matrices the second positive factor retains the entire mixed term

$$\begin{aligned}\mathcal{B}_N^{\text{GCR}} &= I - \tilde{V}_N^* \tilde{K}_N^{-1} \tilde{V}_N \\ &\quad + \tilde{V}_N^* \tilde{K}_N^{-1} \tilde{U}_N (I + \tilde{U}_N^* \tilde{K}_N^{-1} \tilde{U}_N)^{-1} \tilde{U}_N^* \tilde{K}_N^{-1} \tilde{V}_N.\end{aligned}$$

Multiplying the resulting inverse by $\tilde{K}_N + \tilde{U}_N \tilde{U}_N^*$ verifies this equality with the displayed order of every factor.

The note's scalar $d_N^{\text{note}} = \log \det(I + X_N)$ is a logarithmic update, whereas GCR23's d_N is the positive determinant $\mathfrak{d}_N$ in JSR9. Their exact relation is

$$\log \mathfrak{d}_N = d_N^{\text{note}} - \beta_N^{\text{note}}, \quad \beta_N^{\text{note}} = -\log \det(I - Y_N). \quad (\text{JSR10})$$

This also identifies both objects with the original source and relation determinants, as follows.

24 Exact source and relation determinant dictionary

Write H_N^f for the coefficient Gram in $|f|^2 d\sigma$. The complete Gamma monic norms give the positive source ratio

$$D_{l,N} = \frac{\det H_N^f}{\det H_N^\sigma} = \prod_{a=0}^N \prod_{r=0}^{2l-1} (a + r + \tfrac{1}{2}). \quad (\text{JSR11})$$

Indeed the ratio of the monic norms at index a is $\gamma_{k,a}/(\beta_k \gamma_a) = \Gamma(a + 2l + 1/2)/\Gamma(a + 1/2)$, the displayed finite product. Products of all monic norms equal the coefficient determinant because the monic triangular change of frame has determinant one. For $r_N = N + 1 - q$ define the full relation ratio

$$R_N = \frac{\det \text{Gram}_\sigma(f\chi, f\chi S, \dots, f\chi S^{r_N-1})}{\det \text{Gram}_\sigma(\chi, \chi S, \dots, \chi S^{r_N-1})}. \quad (\text{JSR12})$$

At $r_N = 0$ both empty determinants are one. The coefficient frame $(1, \dots, S^{q-1}, \chi, \dots, \chi S^{N-q})$ has determinant one. Block elimination along the entire relation kernel gives $\det H_N = \det H_{\text{rel}} \det G_N$. Applying this twice yields $\det G_N^f / \det G_N^\sigma = D_{l,N} / R_N$. On the other hand JSR6 and JSR9 give that ratio as $|\det M_f|^2 / \mathfrak{d}_N$. Consequently the exact dictionary is

$$\mathfrak{d}_N = |\det M_f|^2 \frac{R_N}{D_{l,N}}, \quad d_N^{\text{note}} - \beta_N^{\text{note}} = 2 \log |\det M_f| + \log R_N - \log D_{l,N}. \quad (\text{JSR13})$$

This is an equality on the original remainder and relation spaces. Every full-unit multiplication is retained by JSR4. For the original boundary $\Lambda : E \rightarrow \text{im } \Lambda$, the product-jet observation is exactly $(0, Y) \mapsto \Lambda E^{-1} U_f^{-1} Y$. It sends $(0, U_f E v)$ to Λv , so the source comparison also retains the actual boundary.

Put $(N_0, N_1, N_2, N_3) = (q-1, q, 2q-1, 2q)$ and $(\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) = (1, 1, -1, -1)$. Define the same four-volume functional for each original quotient metric and retain $\mathfrak{T}_k = F_{0,k} - F_{\sigma,k}$. Only after taking those four metrics and signs do $q \log \beta_k$ and $2 \log |\det M_f|$ cancel. Thus

$$\begin{aligned}\mathcal{H}_k &:= \mathcal{R}_k^0 - \mathcal{R}_k^\sigma = \sum_{a=0}^3 \epsilon_a \log \mathfrak{d}_{N_a} = -\mathfrak{T}_k, \\ \mathcal{H}_k &= \sum_{a=0}^3 \epsilon_a \{ \log \det(I + X_{N_a}) + \log \det(I - Y_{N_a}) \}.\end{aligned} \quad (\text{JSR14})$$

In the precise GSR low-endpoint notation, $R_{q-1} = 1$ and $R_q = \mathfrak{r}_\chi = \frac{\int |f\chi|^2 d\sigma}{\int |\chi|^2 d\sigma}$. Therefore

$$\begin{aligned}C_k^{\text{rel}} &= \log D_{l,q-1} + \log D_{l,q} - \log \mathfrak{r}_\chi - \log D_{l,2q-1} - \log D_{l,2q}, \\ \mathcal{H}_k &= -C_k^{\text{rel}} - \log R_{2q-1} - \log R_{2q}.\end{aligned} \quad (\text{JSR15})$$

The source mass β_k is not C_k^{rel} . Every logarithm is of a positive scalar, and the two low relation dimensions remain exactly zero and one.

25 The finite arithmetic majorant at each original degree

The original arithmetic density is $w_h(y) = |v_h(1/2 + iy)|^2/(2\pi)$, $m_k = w_h^{*k}$. Retain the proved TW/DMR positive constants and the original source determinant deficit

$$\Xi_{k,N} = (N+1) \log A_k - \log \frac{\det H_{k,N}}{\det H_N^\sigma}, \quad A_k = \frac{C_h}{c_\Gamma} k^{p+1} M_\alpha^{k-1}. \quad (\text{JSR16})$$

Here C_h is the original tilted-envelope constant. For the finite bound all required constants are

$$\begin{aligned} p &= 8m - \frac{9}{2}, \quad B = 42 + 8m, \quad M_0 = 21 + 4m, \quad \eta = (p-1)/p, \quad \alpha = \pi/2, \quad g_0 = \log \sqrt{2\pi}, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy, \quad M_1 = \int_{-1}^1 e^{\alpha y} w_h(y) dy, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy, \\ b_h &= \frac{c_h}{C_\Gamma 2^{3B/2}}, \quad C_r = 1 + \frac{\alpha + \log(M_\alpha/\vartheta_h) + B}{\log 2}, \quad C_d = 2C_r + M_0/3, \quad C_L = 1 + C_d, \\ C_E &= 6C_r^2 + (2M_0 + 4)C_d/3, \quad C_s = \left| \log \frac{C_h}{c_\Gamma b_h} + 2 \log M_\alpha \right|, \quad C_{\text{tail}} = \frac{2C_h}{(p-1)M_1}, \\ T_k^{\text{deg}} &= \log(4m(C_L + 2)) + 3 \log(k+2), \quad z_k = \frac{kT_k^{\text{deg}}}{q}, \quad R_k^* = \max(1, z_k^{-1/p}). \end{aligned} \quad (\text{JSR17})$$

The letter M_0 denotes the original DMR exponent, and does not replace the joint source degree $M = N + l$. The Gamma-envelope constants remain $c_\Gamma = \sqrt{2}e^{-7/6}$ and $C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}$. Every constant except $T_k^{\text{deg}}, z_k, R_k^*$ is fixed as k varies. The tilted finite mass M_1 and the un-tilted finite mass ϑ_h remain their separate integrals.

For each $N \in \{q-1, q, 2q-1, 2q\}$ put

$$\begin{aligned} \Omega_{k,N}^{\text{deg}} &= \frac{N+1}{q} \left\{ C_{\text{tail}} \min(2^{1-p}, z_k^\eta) + \frac{(p+1) \log k + C_s}{k} \right\} + \frac{2M_0 N T_k^{\text{deg}}}{kq} \\ &\quad + C_E z_k + C_d \max(z_k, z_k^\eta) + \frac{(g_0 + 2)C_d}{q}, \\ D_k^- &= kq(\Omega_{k,q-1}^{\text{deg}} + \Omega_{k,q}^{\text{deg}}), \quad D_k^+ = kq(\Omega_{k,2q-1}^{\text{deg}} + \Omega_{k,2q}^{\text{deg}}). \end{aligned} \quad (\text{JSR18})$$

This scalar $\Omega_{k,N}^{\text{deg}}$ is not the positive matrix Ω_N in JSR7. Their domains and definitions are explicit. The full finite source statement is

$$0 \leq \Xi_{k,N} \leq kq \Omega_{k,N}^{\text{deg}}. \quad (\text{JSR19})$$

To verify its substitution from the already proved complete DMR source determinant bound, keep the exact factors in ODD7:

$$\begin{aligned} \frac{\Xi_{k,N}}{kq} &\leq \frac{N+1}{q} \left\{ C_{\text{tail}}(1+R)^{1-p} + \frac{(p+1) \log k + C_s}{k} \right\} + \frac{2M_0 N T_k^{\text{deg}}}{kq} \\ &\quad + \frac{kT_k^{\text{deg}}}{q}(C_E + C_d R) + \frac{(g_0 + 2)C_d}{q}, \quad R \geq 1. \end{aligned}$$

At $R = R_k^*$ one has $z_k R_k^* = \max(z_k, z_k^\eta)$ and $(1 + R_k^*)^{1-p} \leq \min(2^{1-p}, z_k^\eta)$. For $z_k \leq 1$ these follow from $R_k^* = z_k^{-1/p}$; for $z_k \geq 1$ they follow from $R_k^* = 1$ and $2^{1-p} \leq 1 \leq z_k^\eta$. This proves JSR19 including the common boundary $z_k = 1$. The complete DMR proof and the unchanged ODD provider are preserved with this source; the inequality is their exact finite degree-dependent receiving substitution.

Write b_N for the original quotient deficit and b_N^K, b_N^B for its fixed original kernel and boundary deficits. The full source/quotient Schur decomposition gives $0 \leq b_N^K, b_N^B \leq b_N \leq \Xi_{k,N}$ and $b_N^K + b_N^B = b_N$. To see the precise maps, for either source use $G_N = (J_N H_N^{-1} J_N^*)^{-1}$, $H_N^K = I_K^* G_N I_K$ and $Q_N^B = (\Lambda G_N^{-1} \Lambda^*)^{-1}$. The common coefficient frame $[I_K, S_*]$ has the same determinant in both sources; its orthogonal lift differs by a kernel column and a triangular matrix of determinant one. Consequently $\det H_N^K \det Q_N^B = |\det[I_K, S_*]|^2 \det G_N$. The quotient Schur factor and its kernel restriction are positive contractions after the same full reference congruence, proving the nonnegative deficit decomposition. Empty factors at rank zero have determinant one.

The original four signs give the sharper finite interval

$$\begin{aligned} a_k^- &= \Xi_{k,q-1} + \Xi_{k,q} \leq D_k^-, & a_k^+ &= \Xi_{k,2q-1} + \Xi_{k,2q} \leq D_k^+, \\ -D_k^- &\leq -a_k^- \leq X_k \leq a_k^+ \leq D_k^+, \\ X_k &\in \{\delta_k^\sigma, \Delta F_K, \Delta F_B\}. \end{aligned} \tag{JSR20}$$

Indeed $X_k = -b_{q-1} - b_q + b_{2q-1} + b_{2q}$, using the corresponding superscript for either original summand. Drop the positive two terms for the lower bound and the negative two for the upper bound, then use JSR19. The original source inclusions into degree $2q$ give $\Xi_{k,N} \leq \Xi_{k,2q}$, hence both $a_k^\pm$ are at most $2\Xi_{k,2q}$. Thus the actual interval using a_k^-, a_k^+ supersedes the note's symmetric interval. The explicit majorants D_k^-, D_k^+ remain outer bounds when smaller exact deficit sums are already known.

For $m \geq 2$ let $C_m = p+1+6M_0 = 32m+245/2$. At an original endpoint with $N/q \rightarrow s \in \{1, 2\}$,

$$\begin{aligned} \frac{k\Omega_{k,N}^{\deg}}{\log k} &\longrightarrow sC_m, & \limsup \frac{\Xi_{k,N}}{q \log k} &\leq sC_m, \\ \frac{D_k^-}{q \log k} &\longrightarrow 64m + 245, & \frac{D_k^+}{q \log k} &\longrightarrow 128m + 490, \\ -(64m + 245) &\leq \liminf \frac{X_k}{q \log k} \leq \limsup \frac{X_k}{q \log k} \leq 128m + 490. \end{aligned} \tag{JSR21}$$

The proof retains $q = (m-1)k^3 + O_m(k^2)$ and $\eta > 1/2$. Since $T_k^{\deg}/\log k \rightarrow 3$, $kz_k^\eta/\log k = O_m(k^{1-2\eta}(\log k)^{\eta-1}) \rightarrow 0$, $kz_k/\log k = O_m(k^{-1})$, and $k/(q \log k) \rightarrow 0$, the two surviving terms in JSR18 have limits $s(p+1)$ and $6M_0s$. This proves the first line; summing the two low and two high endpoints separately proves the rest. For $m=1$ retain $q = (k+1)^2$ and $\eta = 5/7$. The same finite formulas instead give $X_k/(kq) = O_h((\log k/k)^{5/7})$. No $q \log k$ bound in JSR21 is assigned to that branch.

26 The actual finite trace interval receives the asymmetric bounds

Retain the note's common scalar and all eight metrized matrices

$$s_k = 2 + \sum_{a=0}^3 \text{Tr } X_{N_a}, \quad Z_{N_a} = I - (I + X_{N_a})/s_k, \quad Y_{N_a} \quad (0 \leq a \leq 3). \tag{JSR22}$$

Because each eigenvalue of X_N is nonnegative and at most its trace, Z_N has eigenvalues in $[0, 1)$ in the original Ω_N metric. The same assertion for Y_N in its original A_{N+l} metric was proved in JSR7. For any one of these matrices T define

$$b_p(T) = \sum_{n=1}^p \frac{\text{Tr } T^n}{n}, \quad a_p(T) = \text{Tr } T^p, \quad u_p(T) = b_p(T) + \frac{b_{2p}(T) - b_p(T)}{1 - a_p(T)}. \tag{JSR23}$$

At every admitted positive integer order p with $a_p(T) < 1$, $b_p(T) \leq -\log \det(I - T) \leq u_p(T)$. Here is the finite-tail proof: for each eigenvalue $0 \leq x < 1$, let $r_p(x) = \sum_{n>p} x^n/n$. Then $r_{2p}(x) \leq$

$x^p r_p(x)$ by shifting its index. Summing the nonnegative terms gives $\sum r_p \leq b_{2p} - b_p + a_p \sum r_p$, which proves the upper bound. Since every eigenvalue is strictly less than one, $a_p \rightarrow 0$ and $u_p - b_p \rightarrow 0$; in particular every prescribed positive tolerance admits a finite order.

For each endpoint take admitted orders p_a, q_a and put

$$\begin{aligned} L_a &= b_{p_a}(Z_{N_a}) + b_{q_a}(Y_{N_a}), & U_a &= u_{p_a}(Z_{N_a}) + u_{q_a}(Y_{N_a}), \\ \mathcal{L}_k^{\text{note}} &= L_2 + L_3 - U_0 - U_1, & \mathcal{U}_k^{\text{note}} &= U_2 + U_3 - L_0 - L_1. \end{aligned} \quad (\text{JSR24})$$

The four original $l \log s_k$ terms cancel with their four signs, so JSR14 is

$$\mathcal{H}_k = - \sum_a \epsilon_a \sum_{n \geq 1} (\text{Tr } Z_{N_a}^n + \text{Tr } Y_{N_a}^n) / n.$$

Inserting JSR23 with these signs proves $\mathcal{L}_k^{\text{note}} \leq \mathcal{H}_k \leq \mathcal{U}_k^{\text{note}}$. This finite interval uses the same actual entries and metrics as the note, GCR and GSR via JSR8–15.

The exact GSR low endpoints give a further contraction of this particular trace interval. Define their complete sum and the new finite endpoints by

$$\begin{aligned} A_k^{\text{low}} &= 4 \log |\det M_f| - \log D_{l,q-1} - \log D_{l,q} + \log \mathfrak{r}_\chi, \\ \mathcal{L}_k &= A_k^{\text{low}} - 2l \log s_k + L_2 + L_3, & \mathcal{U}_k &= A_k^{\text{low}} - 2l \log s_k + U_2 + U_3, \\ \mathcal{L}_k^{\text{note}} &\leq \mathcal{L}_k \leq \mathcal{H}_k \leq \mathcal{U}_k \leq \mathcal{U}_k^{\text{note}}, \\ \mathcal{U}_k - \mathcal{L}_k &= (U_2 - L_2) + (U_3 - L_3). \end{aligned} \quad (\text{JSR24a})$$

To prove this, put $Q_a = -\log \det(I - Z_{N_a}) - \log \det(I - Y_{N_a})$. Then $\log \mathfrak{d}_{N_a} = l \log s_k - Q_a$ and $L_a \leq Q_a \leq U_a$. JSR13 with $R_{q-1} = 1$, $R_q = \mathfrak{r}_\chi$ proves $\log \mathfrak{d}_{q-1} + \log \mathfrak{d}_q = A_k^{\text{low}}$. Consequently $\mathcal{H}_k = A_k^{\text{low}} - 2l \log s_k + Q_2 + Q_3$. This proves its new endpoints. Also $Q_0 + Q_1 = 2l \log s_k - A_k^{\text{low}}$ lies between $L_0 + L_1$ and $U_0 + U_1$, which proves both outer containments and the exact width formula. Thus JSR24a is the substitution of the accepted GSR3–6 low-endpoint identities through the exact maps JSR8–13 into the note's actual trace interval. It is the same original determinant in those three presentations.

Every low-endpoint quantity here has a finite expression on the original polynomial source. In particular, let a_χ be the full degree- q coefficient column of χ , let F_q be multiplication by f from $\mathcal{P}_q$ into $\mathcal{P}_{q+l}$, and let P_d have as columns the monic polynomials $p_0, \dots, p_d$ in the original S coordinate. Then

$$H_d^\sigma = P_d^{-*} \text{diag}(\gamma_0, \dots, \gamma_d) P_d^{-1}, \quad \mathfrak{r}_\chi = \frac{a_\chi^* F_q^* H_{q+l}^\sigma F_q a_\chi}{a_\chi^* H_q^\sigma a_\chi}. \quad (\text{JSR24b})$$

The first identity follows from monic orthogonality and the second from the exact coefficient multiplication map. Both are finite, positive, and retain the full polynomial χ with all multiplicities. Thus substituting these low endpoints introduces no new unknown source measure.

The updated actual arithmetic receiver is therefore

$$\boxed{\begin{aligned} \mathcal{B}_k^0 + \mathcal{L}_k - a_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 + \mathcal{U}_k + a_k^+, \\ \mathcal{B}_k^0 + \mathcal{L}_k - D_k^- &\leq \mathcal{B}_k^{\text{ar}} \leq \mathcal{B}_k^0 + \mathcal{U}_k + D_k^+. \end{aligned}} \quad (\text{JSR25})$$

Indeed the unchanged TWA equality is exactly $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^0 + \mathcal{H}_k + \delta_k^\sigma$; add its two proved signed intervals. The first line is contained in the note's interval with the same trace endpoints and $\pm 2\Xi_{k,2q}$. The second line evaluates the arithmetic allowances by their complete finite source constants. Equivalently these bounds apply to $\Delta_k^\Gamma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^0$. In particular

$$\Delta_k^\Gamma - \mathcal{H}_k = \Delta_k^\Gamma + \mathfrak{T}_k = \delta_k^\sigma. \quad (\text{JSR26})$$

The constants in JSR21 apply to this exact difference, not to Δ_k^Γ with its Gamma return omitted.

There is an entirely finite order choice that propagates the constants to the width of the actual receiver. For each of the four high-endpoint matrices choose the least positive integer p satisfying $a_p(T) < 1$ and $u_p(T) - b_p(T) \leq 1/4$. The convergence proved after JSR23 proves that each least order exists. Then

$$\begin{aligned}
0 &\leq \mathcal{U}_k - \mathcal{L}_k = \sum_{a=2}^3 (U_a - L_a) \leq 1, \\
\limsup \frac{(\mathcal{U}_k + a_k^+) - (\mathcal{L}_k - a_k^-)}{q \log k} &\leq 192m + 735 \quad (m \geq 2), \\
\lim \frac{(\mathcal{U}_k + D_k^+) - (\mathcal{L}_k - D_k^-)}{q \log k} &= 192m + 735 \quad (m \geq 2).
\end{aligned} \tag{JSR27}$$

The finite widths are bounded by $1 + a_k^- + a_k^+$ and $1 + D_k^- + D_k^+$ respectively. The exact width identity in the first line and the two limits in JSR21 prove both claims. This order choice requires actual matrix traces; it inserts no unproved uniform bound for them as the packet grows. When numerical entries are used, their certified errors must also enter the finite intervals before these claims are invoked.

Finally the original HC residual retains its constant, sign and nonnegative constraint:

$$\begin{aligned}
\mathcal{R}_k &= \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k) \geq 0, \\
\max\{0, \mathcal{B}_k^0 + \mathcal{L}_k - D_k^- - 4q \log(D_h k)\} &\leq \mathcal{R}_k \\
&\leq \mathcal{B}_k^0 + \mathcal{U}_k + D_k^+ - 4q \log(D_h k).
\end{aligned} \tag{JSR28}$$

Subtract the same original threshold from JSR25 and intersect its lower endpoint with the proved nonnegative constraint. Both the complete arithmetic source and the entire signed joint-Schur return remain in this formula. It makes no assertion of an RH endpoint or of a leading value for the signed Gamma term.

Exact source dependencies. This calculation uses the received *Tau Gamma Joint Schur Return* note, the complete GCR1–44 source, the complete GSR1–33/JQR1–39 source, ODD1–14, and the accepted full HMR successor including HMR44a–44c. The companion JSON records their complete source hashes and unchanged copies. The new argument above proves the coordinate dictionary, the finite signed receiving substitutions and their original-degree constants; it does not reattribute the complete inherited DMR determinant proof to this note.

The signed resolvent on the original joint-Schur return

This calculation supplies the direct coefficient maps, positive metric, traceless observable, finite polynomial, and integrated error for the joint-Schur return. Its finite stopping constants are functions of the specified packet and degree. Every matrix below is constructed from the original sigma source and its full remainder map. Equation labels JRA1–JRA29 are local to this complete calculation.

Original objects and full coefficient maps. Retain $0 < \delta < 1/2$, $\gamma > 2$, integer $m \geq 1$, and

$$k = 4l + 1, \quad l \geq 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = (k + 1)^2 e, \\ \lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^e, \quad E = \mathbb{C}[S]/(\chi). \quad (\text{JRA1})$$

The nodes are distinct: equality forces equality of their real and imaginary parts, hence equality of a, b . None is real because k is odd. Put $\mathcal{P}_N = \{P \in \mathbb{C}[S] : \deg P \leq N\}$ with its given sigma form

$$\langle P, Q \rangle_\sigma = \int_{\mathbb{R}} \overline{P(c + iy)} Q(c + iy) \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad H_N^\sigma = (\langle S^i, S^j \rangle_\sigma)_{i,j=0}^N. \quad (\text{JRA2})$$

These are the finite positive source forms of the incoming joint-Schur construction. Positivity follows directly since their density is positive and a nonzero polynomial cannot vanish on the entire line; finiteness is the Gamma-moment property of that original source. We retain its mass $c_\sigma = \sqrt{2\pi}$ in every H_N^σ .

The coefficient map $J_N : \mathcal{P}_N \rightarrow E$ is monic remainder modulo χ , in the basis $1, S, \dots, S^{q-1}$. It is onto for $N \geq q - 1$. For $\alpha_j = c + 2j + 1/2$ and $f(S) = \prod_{j=0}^{l-1} (S - \alpha_j)$ define $Z_M : \mathcal{P}_M \rightarrow \mathbb{C}^l$ by $(Z_M P)_j = P(\alpha_j)$. For $M = N + l$ the joint map is

$$\mathcal{J}_M = \begin{pmatrix} J_M \\ Z_M \end{pmatrix} : \mathcal{P}_M \longrightarrow E \oplus \mathbb{C}^l, \quad \mathcal{J}_M (H_M^\sigma)^{-1} \mathcal{J}_M^* = \begin{pmatrix} K_M & C_M \\ C_M^* & A_M \end{pmatrix}. \quad (\text{JRA3})$$

To prove surjectivity, χ and f are coprime by JRA1 and the reality of α_j . Their Chinese remainder map is onto and has representatives of degree at most $q + l - 1 \leq M$. Its two factors are E and the evaluation algebra at the distinct α_j . Thus the block covariance in JRA3 is positive definite: its quadratic form on $v \neq 0$ is the strictly positive $(H_M^\sigma)^{-1}$ form on $\mathcal{J}_M^* v \neq 0$. Consequently K_M, A_M and both Schur complements

$$G_M^\sigma = K_M^{-1}, \quad S_M = K_M - C_M A_M^{-1} C_M^*, \quad L_M = A_M - C_M^* G_M^\sigma C_M \quad (\text{JRA4})$$

are positive definite. For example, minimizing the JRA3 form over the first coordinate with second coordinate z gives $z^* L_M z > 0$; minimizing over the second proves the assertion for S_M .

Let p_n be the monic orthogonal polynomials in this same sigma form, with squared norms $\omega_n^\sigma > 0$. Orthogonal decomposition of $\mathcal{P}_{N+l}$ into $\mathcal{P}_N$ and its monic tail gives

$$U_N = [Jp_{N+1}, \dots, Jp_{N+l}] : \mathbb{C}^l \rightarrow E, \quad \Omega_N = \text{diag}(\omega_{N+1}^\sigma, \dots, \omega_{N+l}^\sigma), \\ X_N = \Omega_N^{-1} U_N^* G_N^\sigma U_N, \quad Y_N = A_{N+l}^{-1} C_{N+l}^* G_{N+l}^\sigma C_{N+l}. \quad (\text{JRA5})$$

Thus $\Omega_N X_N = U_N^* G_N^\sigma U_N \geq 0$ and $A_{N+l} Y_N = C_{N+l}^* G_{N+l}^\sigma C_{N+l} \geq 0$. Moreover $A_{N+l}(I - Y_N) = L_{N+l} > 0$. The original metrics are Ω_N on the first coefficient space and A_{N+l} on the second. It follows, by conjugation with their positive square roots, that X_N has nonnegative real eigenvalues and Y_N has real eigenvalues in $[0, 1)$. No cross covariance has been removed.

In raw derivative coordinates the remainder coordinate map is

$$\begin{aligned} (\mathbf{E}_\chi)_{(\lambda,r),n} &= \begin{cases} n!\lambda^{n-r}/(n-r)!, & n \geq r, \\ 0, & n < r, \end{cases} & \mathbf{E}_\chi M_f \mathbf{E}_\chi^{-1} &= \bigoplus_\lambda F_\lambda, \\ (F_\lambda)_{r,s} &= \begin{cases} \binom{r}{s} f^{(r-s)}(\lambda), & r \geq s, \\ 0, & r < s. \end{cases} \end{aligned} \quad (\text{JRA6})$$

All $0 \leq r < e$ occur, and entries between different nodes are zero. The equality is Leibniz's formula. The inverse is Hermite interpolation, since vanishing of all these derivatives is divisibility by χ and the domain has dimension q . If an earlier file uses divided derivatives, its explicit map to JRA6 is $D = \text{diag}_{\lambda,r}(r!)$: raw jets equal D times divided jets, and the multiplication matrices obey $F_{\text{raw}} = D F_{\text{div}} D^{-1}$. This does not change the original remainder map or drop a higher jet.

The determinant character representing the actual return. Write $V_N^\sigma = \det G_N^\sigma$ and let G_N^0 denote the tensor-Gamma quotient form in the original remainder coordinates. Gamma recurrence on $k = 4l + 1$ gives exactly

$$dm_k^0 = C_k |f(c + iy)|^2 d\sigma(y), \quad C_k = \frac{(\sqrt{2\pi})^k}{2^{1-k/2} \Gamma(k/2)} 4^{-l}. \quad (\text{JRA7})$$

Indeed the holomorphic ratio is $(i/2)^l \prod_{j=0}^{l-1} (y - i(2j + 1/2))$; taking its real-line modulus gives JRA7, including the factor 4^{-l} . Multiplication by f maps $\mathcal{P}_N$ isomorphically onto $\ker Z_{N+l}$, sends its remainder v to $M_f v$, and changes squared norms by C_k . Minimizing the sigma norm under the joint constraint $(M_f v, 0)$, then inverting the positive block matrix JRA3, proves

$$G_N^0 = C_k M_f^* S_{N+l}^{-1} M_f, \quad \log V_N^0 = q \log C_k + 2 \log |\det M_f| + \log V_{N+l}^\sigma - \log \det(I - Y_N). \quad (\text{JRA8})$$

For the determinant equality, Schur elimination gives $\det S_M \det A_M = \det K_M \det L_M$; divide by $\det A_M \det K_M$ and use $I - Y_N = A_M^{-1} L_M$. Here M_f is invertible because f, χ are coprime.

The tail decomposition used in JRA5 gives $K_{N+l} = K_N + U_N \Omega_N^{-1} U_N^*$. Taking determinants and using $\det(I + AB) = \det(I + BA)$ gives

$$\log V_N^\sigma - \log V_{N+l}^\sigma = \log \det(I + X_N). \quad (\text{JRA9})$$

The determinant identity follows, for instance, by computing the determinant of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ by either Schur complement. Set

$$(N_0, N_1, N_2, N_3) = (q - 1, q, 2q - 1, 2q), \quad (\varepsilon_0, \varepsilon_1, \varepsilon_2, \varepsilon_3) = (1, 1, -1, -1). \quad (\text{JRA10})$$

The signed quotient-volume difference is therefore

$$\mathcal{R} := \sum_{r=0}^3 \varepsilon_r (\log V_{N_r}^\sigma - \log V_{N_r}^0) = \sum_{r=0}^3 \varepsilon_r \{ \log \det(I + X_{N_r}) + \log \det(I - Y_{N_r}) \}. \quad (\text{JRA11})$$

The $q \log C_k + 2 \log |\det M_f|$ terms in JRA8 cancel because $\sum_r \varepsilon_r = 0$, after their four original appearances have been retained. In the incoming notation, $\mathcal{R} = R_k^0 - R_k^\sigma$.

Define the ordered direct sum and its metric by

$$\begin{aligned} \mathcal{V} &= \bigoplus_{r=0}^3 (\mathcal{V}_{r,+} \oplus \mathcal{V}_{r,-}), & \mathcal{V}_{r,\pm} &= \mathbb{C}^l, & n &= 8l, \\ \mathcal{M} &= \text{diag}_{r=0}^3 (\Omega_{N_r}, A_{N_r+l}), \\ \mathcal{B} &= \text{diag}_{r=0}^3 (I + X_{N_r}, I - Y_{N_r}), & Q &= \text{diag}_{r=0}^3 (\varepsilon_r I_l, \varepsilon_r I_l). \end{aligned} \quad (\text{JRA12})$$

Both Q and $\mathcal{B}$ are $\mathcal{M}$ -self-adjoint, $\mathcal{M}\mathcal{B} > 0$, and direct multiplication gives

$$Q^2 = I, \quad \text{Tr } Q = 2l \sum_r \varepsilon_r = 0, \quad \text{Tr } Q^2 = n = 8l. \quad (\text{JRA13})$$

The exact scalar map is the determinant character

$$\Delta : \prod_{r,\pm} \text{GL}(\mathcal{V}_{r,\pm}) \rightarrow \mathbb{C}^\times, \quad (T_{r,\pm}) \mapsto \prod_{r=0}^3 \{\det T_{r,+} \det T_{r,-}\}^{\varepsilon_r}. \quad (\text{JRA14})$$

It is multiplicative because each determinant is. Equivalently it is the action on $\bigotimes_r (\det \mathcal{V}_{r,+} \otimes \det \mathcal{V}_{r,-})^{\varepsilon_r}$, with exponent -1 meaning the dual line. On $\mathcal{B}$, its value is positive and $\log \Delta(\mathcal{B}) = \mathcal{R} = \text{Tr}(Q \log \mathcal{B})$ by JRA11. This is the typed map connecting the new eight coefficient summands to the four original quotient determinants; it does not assert that Q is the earlier four-projector contrast on the much larger source.

A complete simultaneous spectral frame. Construct W as the direct sum of the positive square roots of the eight blocks Ω_{N_r}, A_{N_r+l} of $\mathcal{M}$. Then $W^*W = \mathcal{M}$, W^{-1} exists, and $\widehat{B} = W\mathcal{B}W^{-1}$ is Hermitian positive. Construct an orthonormal eigenbasis for $\widehat{B}$ separately in its eight invariant blocks. For completeness, a Hermitian matrix has a real eigenvalue by the fundamental theorem of algebra and $\lambda v^*v = v^*Hv = \bar{v}^*\bar{H}\bar{v}$; the orthogonal complement of an eigenvector is invariant because $v^*Hw = (Hv)^*w$. Induction constructs an orthonormal eigenbasis, and positivity makes all eigenvalues positive. Apply the same construction to the positive matrix $\mathcal{M}$ to construct its positive square root, taking the positive square root of each eigenvalue. Thus both constructions and both inverse laws follow. If U is the resulting block unitary matrix, put $F = U^*W$. Then, including every eigenvalue with multiplicity,

$$\begin{aligned} F^*F &= \mathcal{M}, & FF^{-1} &= F^{-1}F = I, \\ FB F^{-1} &= \text{diag}(b_1, \dots, b_n), & b_i &> 0, \\ FQF^{-1} &= \text{diag}(\epsilon_1, \dots, \epsilon_n). \end{aligned} \quad (\text{JRA15})$$

Here each ε_r occurs $2l$ times. The last equality follows because Q is scalar on each of the eight blocks, so the constructed block isometry commutes with its signs. Conjugation preserves products, powers, inverses, and traces by direct multiplication and cyclicity.

Finite resolvent with explicit positive constants. Constants requiring only the eight original matrices are

$$\begin{aligned} \mu &= \min(1, \min_{0 \leq r \leq 3} \det(I - Y_{N_r})) > 0, & \nu &= 1 + \sum_{r=0}^3 \text{Tr } X_{N_r} \geq 1, \\ d &= (\mu + \nu)/2 > 0, & \theta &= (\nu - \mu)/(\nu + \mu) \in [0, 1), \\ K_0 &= \mu^{-2} \text{Tr}(\mathcal{B} - I)^2 = \mu^{-2} \sum_{r=0}^3 \text{Tr}(X_{N_r}^2 + Y_{N_r}^2) \geq 0. \end{aligned} \quad (\text{JRA16})$$

Every eigenvalue of $I - Y_N$ lies in $(0, 1]$. Its determinant, the product of all eigenvalues, is at most each one; hence $b_i \geq \mu$ on these blocks. On the other blocks $b_i \geq 1 \geq \mu$ and $b_i \leq 1 + \text{Tr } X_{N_r} \leq \nu$. The minus blocks also have $b_i \leq 1 \leq \nu$. This proves $\mu \leq b_i \leq \nu$ and all claims in JRA16, including $\mu \leq 1$ and $\mu \leq \nu$.

For $0 \leq t \leq 1$ set

$$\begin{aligned} B_t &= (1 - t)I + t\mathcal{B}, & a_t &= 1 + t(d - 1) > 0, \\ X_t^{\text{res}} &= B_t^{-1}(\mathcal{B} - I), & H_t &= I - B_t/a_t = t(dI - \mathcal{B})/a_t, \end{aligned}$$

$$T_t = (\mathcal{B} - I)/a_t, \quad S_L(t) = \sum_{j=0}^L H_t^j T_t \quad (L \geq 0). \quad (\text{JRA17})$$

All these endomorphisms are self-adjoint in the original $\mathcal{M}$ and commute with each other, being real rational functions of $\mathcal{B}$ with positive denominators. In JRA15 their diagonal values are

$$\lambda_i(t) = 1 + t(b_i - 1), \quad \gamma_i(t) = \frac{b_i - 1}{\lambda_i(t)}, \quad h_i(t) = \frac{t(d - b_i)}{a_t}. \quad (\text{JRA18})$$

These equations explicitly supply the simultaneous diagonalization required by the PR31 spectral-residual theorem. They follow by applying F to each factor of JRA17; the diagonal values have not been assigned to an unrelated replacement matrix. We have $\lambda_i(t) \geq \min(1, b_i) \geq \mu$ and $|\gamma_i(t)| \leq |b_i - 1|/\mu$. Since $|d - b_i| \leq (\nu - \mu)/2$ and $t/a_t \leq 1/d$ (equivalently $td \leq 1 + t(d - 1)$),

$$|h_i(t)| \leq \theta < 1, \quad \sum_i \gamma_i(t)^2 \leq K_0. \quad (\text{JRA19})$$

Direct multiplication gives $(I - H_t)X_t^{\text{res}} = T_t$. The finite telescoping identity $\left(\sum_{j=0}^L H_t^j\right)(I - H_t) = I - H_t^{L+1}$, obtained by cancelling every intermediate power, therefore proves

$$R_L(t) := X_t^{\text{res}} - S_L(t) = H_t^{L+1} X_t^{\text{res}}, \quad E_L(t) := R_L(t) - \frac{\text{Tr } R_L(t)}{n} I. \quad (\text{JRA20})$$

The constant term in S_L and the exponent $L + 1$ are both retained.

The actual centered pointwise error and its integral. Under F , the eigenvalues of $R_L(t)$ are $\delta_i(t) = h_i(t)^{L+1} \gamma_i(t)$. Thus its trace is real and

$$\begin{aligned} V_L(t) &:= \text{Tr } E_L(t)^2 = \sum_i \delta_i(t)^2 - \frac{(\sum_i \delta_i(t))^2}{n} \\ &= \sum_i \left(\delta_i(t) - \frac{1}{n} \sum_j \delta_j(t) \right)^2 \geq 0, \quad V_L(t) \leq \theta^{2L+2} K_0. \end{aligned} \quad (\text{JRA21})$$

Both equalities follow by expanding the square, and the bound follows from JRA19 after dropping its nonnegative subtracted scalar square. Scalar centering retains the signed pairing because $\text{Tr } Q = 0$: $\text{Tr}(R_L Q) = \text{Tr}(E_L Q)$. For Hermitian matrices A, C the entrywise Cauchy–Schwarz inequality gives $|\text{Tr}(AC)| \leq (\text{Tr } A^2 \text{Tr } C^2)^{1/2}$, since $\text{Tr } A^2 = \sum_{i,j} |A_{ij}|^2$ and similarly for C . Apply this to $FE_L F^{-1}$ and FQF^{-1} ; trace invariance and JRA13 give the full original-metric error

$$|\text{Tr}(QX_t^{\text{res}}) - \text{Tr}(QS_L(t))| \leq \sqrt{8l V_L(t)} \leq \theta^{L+1} \sqrt{8l K_0}. \quad (\text{JRA22})$$

This reproduces the centered PR31 estimate on the specified joint-Schur matrices, with its exact trace $8l$. It uses the generic signed trace and spectral-residual results; it does not substitute $8l$ for the different source-projector estimate $8q - 6$.

For each i , elementary integration yields $\log b_i = \int_0^1 (b_i - 1)/(1 + t(b_i - 1)) dt$, also when $b_i = 1$. All denominators are at least $\mu > 0$, so these finite sums are continuous and can be integrated term by term. JRA11, JRA15 and JRA18 prove

$$\begin{aligned} \mathcal{R} &= \int_0^1 \text{Tr}(QX_t^{\text{res}}) dt, \quad \mathcal{C}_L = \int_0^1 \text{Tr}(QS_L(t)) dt, \\ |\mathcal{R} - \mathcal{C}_L| &\leq \int_0^1 \sqrt{8l V_L(t)} dt \leq \theta^{L+1} \sqrt{8l K_0}. \end{aligned} \quad (\text{JRA23})$$

The inequality is integration of JRA22 on a unit interval, not a claim that a pointwise bound automatically has been integrated.

The finite center itself has a closed expression in the original matrix coefficients. Put $I_j(d) = \int_0^1 t^j / a_t^{j+1} dt$. Then

$$\begin{aligned} \mathcal{C}_L &= \sum_{j=0}^L I_j(d) \operatorname{Tr}(Q(dI - \mathcal{B})^j (\mathcal{B} - I)), \\ I_j(1) &= \frac{1}{j+1}, \\ I_j(d) &= \frac{\log d + \sum_{a=1}^j (-1)^a \binom{j}{a} \frac{1-d^{-a}}{a}}{(d-1)^{j+1}} \quad (d \neq 1). \end{aligned} \quad (\text{JRA24})$$

For $d \neq 1$, substitute $u = 1 + (d-1)t$; the integrand becomes $(d-1)^{-j-1}(u-1)^j u^{-j-1} du$ on the oriented interval from 1 to d . Expand $(u-1)^j$ and integrate u^{-1} and u^{-a-1} to obtain JRA24, including the case $0 < d < 1$ with its orientation. For $d = 1$ the integrand is t^j . Thus no unresolved matrix logarithm or path integral is needed to compute the finite center. Every trace in it is the signed sum of eight actual coefficient-matrix traces.

For $\eta > 0$, put $A = \sqrt{8lK_0}$. If $A = 0$, then JRA16 and positivity of the eigenvalue squares force $\mathcal{B} = I$, so $\mathcal{R} = 0$ and $\mathcal{C}_0 = 0$. If $\theta = 0$, JRA16 forces $\mu = \nu = 1$, again $\mathcal{B} = I$. Otherwise, with $A > 0$ and $0 < \theta < 1$, the explicit integer

$$L_\eta = \max \left(0, \left\lceil \frac{\log(A/\eta)}{-\log \theta} \right\rceil \right) \quad \text{satisfies} \quad |\mathcal{R} - \mathcal{C}_{L_\eta}| \leq \eta. \quad (\text{JRA25})$$

Indeed, if $A \leq \eta$ every $L \geq 0$ works. Otherwise $L_\eta \geq \log(A/\eta)/(-\log \theta)$ and $\theta^{L_\eta+1} A \leq \theta^{L_\eta} A \leq \eta$. This proves an executable finite stopping rule for the given data.

Exact replacement and support maps. Section 6 of the incoming note gives its positive-tail interval $[L_{\text{tail}}, U_{\text{tail}}]$ for precisely JRA11. The present completed calculation supplies the simultaneously valid intersection

$$\mathcal{R} \in [L_{\text{tail}}, U_{\text{tail}}] \cap [\mathcal{C}_L - \theta^{L+1} \sqrt{8lK_0}, \mathcal{C}_L + \theta^{L+1} \sqrt{8lK_0}]. \quad (\text{JRA26})$$

The intersection contains $\mathcal{R}$ because each interval does. Unlike the positive-tail lower/upper sums, JRA24 forms its signed center before the single centered error is bounded. This is an additional finite certificate; no claim that its radius always beats the earlier interval is needed or made.

The relation-valued section correction remains the same map. Let B_N be the columns of multiplication by χ into $\mathcal{P}_N$, and $R_N^\sigma : E \rightarrow \mathcal{P}_N$ the sigma minimal section. For the tensor-Gamma form H_N^0 , minimization over its relation fibre gives the actual section and exact difference

$$R_N^0 = R_N^\sigma - B_N (B_N^* H_N^0 B_N)^{-1} B_N^* H_N^0 R_N^\sigma, \quad R_N^0 - R_N^\sigma \in \operatorname{Hom}(E, \chi \mathcal{P}_{N-q}). \quad (\text{JRA27})$$

To prove the formula, write every representative of v as $R_N^\sigma v + B_N z$, expand its positive quadratic form, and complete the square in z . B_N has independent columns, so its Gram is positive definite. At $N = q - 1$ it is the zero-dimensional matrix and the correction is zero. The weighted cyclic inclusion obeys

$$\eta M_f = M_{f(\sum s_i) \eta}, \quad \eta[P] = v_h^{\otimes k} P(\sum s_i) 1. \quad (\text{JRA28})$$

This equality is multiplication of commuting elements in the original tensor algebra; all full unit factors and nilpotents remain. Since f is coprime to χ , its Bezout inverse on E intertwines the same way on the image of η . Thus the determinant calculation transports the same source classes through explicitly invertible coefficient maps.

On any fixed retained face and outer label ω , every coefficient map T above acts by $(\omega, x) \mapsto (\omega, Tx)$, and external absence maps to external absence. The represented zero $(\omega, 0)$ stays at that label. Composition is exact since $(\omega, T_2(T_1x)) = (\omega, (T_2T_1)x)$. These statements are maps of labelled coefficient sets; they do not assign extra multiplicative structure to an arbitrary linear map. Neither the direct sum nor the determinant character deletes a present empty face or identifies it with τ .

Finally, for the existing arithmetic correction $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^0 + \mathcal{R} + \delta_k^\sigma$, the proven interval $\delta_k^\sigma \in [a_k, b_k]$ for the arithmetic correction propagates literally:

$$\mathcal{B}_k^{\text{ar}} \in [\mathcal{B}_k^0 + \mathcal{C}_L + a_k - \theta^{L+1}\sqrt{8lK_0}, \mathcal{B}_k^0 + \mathcal{C}_L + b_k + \theta^{L+1}\sqrt{8lK_0}]. \quad (\text{JRA29})$$

This last implication is addition of the two proved intervals; the arithmetic correction theorem itself remains in its original proof files. Every new bound through JRA25 is proved here for the actual joint-Schur matrices. A degree-dependent choice of μ, ν, K_0 has not been replaced by a degree-independent constant. The next analytic calculation is their behaviour, and that of the signed center, along the original growing family.

Formal-source provenance. The generic results used and reproved above are

```
SplitZero.ResolventSeries.residual_exact,
SplitZero.SignedTraceEnclosure.signed_interval, and
SplitZero.SpectralResidual.signed_error
```

at PR31 head b4060b25c21f1a49a5e7730e9aff76d4c93c820d. The source files and full theorem-scope review accompany this TeX. JRA15–JRA21 construct the required spectral equations rather than assuming them as unnamed inputs. JRA23–JRA25 add the path integration and its closed finite center. This application has a written proof; it is not represented as an already compiled Lean declaration.

The complete degree-36 rational calibration

27 Complete degree-36 rational calibration

This is the parameter calibration $k = 5$, $l = 1$, $m = 1$, $q = 36$, $\delta = 1/4$, $\gamma = 3$. These numbers are not asserted to be zeta-zero coordinates. Every original quartet-sum factor is present. The four original cutoffs are 35, 36, 71, 72 with signs $+$, $+$, $-$, $-$.

All X and Y values, their positive Schur gaps, the complete centered packet polynomial, original relation determinants and outward rational logarithm bounds are saved in `EXACT_CALIBRATION.json`. The execution took 0.585751699982211 seconds, excluding report writing; the stages are recorded in `execution.log`. Four full 36-by-36 determinant updates, four independent original relation-Gram returns and four independent full-source Gram products passed. The free A formula passed at all 74 degrees from 0 through 73. The initial array-length guard caught an off-by-one construction bound before any calibration was run; that guard log is preserved and the corrected complete run is the reported one.

The X and gap columns below are approximate displays of the saved exact rationals. Each logarithm interval is a certified outward decimal enlargement of the saved rational enclosure.

N	X_N (display)	$1 - Y_N$ (display)
35	611875.979208	$2.75342492301 \cdot 10^{-26}$
36	1076590.53366	$2.71699841706 \cdot 10^{-26}$
71	25.2901832744	$7.81967451353 \cdot 10^{-16}$
72	33.7860627273	$7.7670095311 \cdot 10^{-16}$

The following columns enclose $\log((1 + X_N)(1 - Y_N))$.

N	Certified lower endpoint	Certified upper endpoint
35	-45.53008032759609368921746133677353312	-45.53008032759609368921746133677353311
36	-44.97837404888811850068746740854904506	-44.97837404888811850068746740854904505
71	-31.51552294654730871528707229105220838	-31.51552294654730871528707229105220837
72	-31.24225946112371394239146781416846352	-31.24225946112371394239146781416846351

The complete signed Schur logarithm $R^0 - R^\sigma$ lies between

$$\begin{aligned} & -27.75067196881318953222638864010190629, \\ & -27.75067196881318953222638864010190628. \end{aligned}$$

The saved rational interval has width $1.506240906334363 \cdot 10^{-39}$. The separately retained free-boundary signed logarithm lies between

$$\begin{aligned} & 0.67627639363778630655740630247752650, \\ & 0.67627639363778630655740630247752651. \end{aligned}$$

27.1 Exact receiving and sign identities

At each endpoint the computation proves exactly

$$\frac{D_{1,N}}{R_N}(1 + X_N)(1 - Y_N) = \chi(1/2)^2.$$

Here R_N is formed independently from the complete original relation columns χx^j , including relation ranks 0, 1, 36, 37; both empty determinants at the first endpoint are one. The translation

from χS^j to χx^j is the full binomial triangular matrix with determinant one, so its determinant ratio is unchanged. The source product

$$D_{1,N} = \prod_{n=0}^N (n + 1/2)(n + 3/2)$$

is also independently recovered by determinants of the complete original f -Gram parity blocks. Their diagonal entries are $d_{n+1} + (1/4)d_n + a_n^2 d_{n-1}$, off-diagonal entries at distance two are $-d_{n+2}$, and $a_n = n(n - 1/2)$. These follow directly from

$$f p_n = p_{n+1} - a_n p_{n-1} - p_n/2$$

in the original orthogonal polynomial frame.

Consequently the four-signed Schur product is exactly reciprocal to GSR's signed source/relation product. NOTE computes $R^0 - R^\sigma$; GSR computes the opposite signed object T_k . No sign convention is silently changed.

The actual mass is retained as

$$K = c_\sigma^{-1} \hat{K}, \quad C = c_\sigma^{-1} \hat{C}, \quad A = c_\sigma^{-1} \hat{A}, \quad \omega_n = c_\sigma d_n, \\ c_\sigma = \sqrt{2\pi}.$$

The c_σ factors cancel in each paired X and Y expression. No source probability measure is substituted. Positive d_n and the first 36 monic remainder columns prove $\hat{K}$ positive definite; the exact positive values of $1 - Y$ prove the positive joint Schur complements.

27.2 Rigorous logarithm evaluation

For each positive exact rational r , choose the exact integer e so that $r = 2^e s$ and $1 \leq s < 2$. Bound s outward by consecutive dyadic rationals of denominator 2^{128} . For either endpoint use $t = (s - 1)/(s + 1)$, so $0 \leq t \leq 1/3$. The first 96 terms of

$$2 \sum_{j \geq 0} \frac{t^{2j+1}}{2j+1}$$

give the lower logarithm bound. The remaining positive tail is at most

$$\frac{2t^{193}}{193(1 - t^2)},$$

giving the upper bound. Apply the same enclosure to $\log 2$, multiply by e with the appropriate direction when $e < 0$, and add. Every step uses exact rational arithmetic. This gives the saved finite bounds even for the small original Schur gaps, without a slowly converging series directly at Y close to one.